Figure 3 Gate

Two Framings of the Single-Deletion Multimodal Figure, as Mockups

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October 11, 2026

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The morphology SI figure, as a sketch to run experiments against ✗

Measured. Panels a, b, c and f can be drawn now from committed result files (tables 1 to 3, and the v20 readout). **Not measured.** Panel d (a measured modality revealed, paired against the permuted and the genotype-only controls) is in flight on `cabbi`; panel e (per feature) has no dump behind it; panel g (the 1,011 isolates) is planned and likely not in time.

Every arm here is transductive in the sense the figure needs. The revealed modality is a gene-level measurement of the *same* strain, present on every training, validation and test strain of the arm (`require_modalities` keeps only genotypes carrying both labels, table 1), and it enters through the observed-label channel on every step with nothing masked, so the model is applied to a strain whose other modality is known. Morphology itself can be a target and not an input: the observed-label channel takes per-gene values, and CalMorph is one vector per cell population. The genotype-only arms (`M_cls`, `MP_cls`) are the inductive controls on exactly the same strains.

What the sketch asks of the runs. Panel d is the figure's claim and its only open measurement: twelve paired differences per contrast, scored on the window rule, with the ridge from the same measured modality as the bar (0.280 from the proteome, 0.324 from expression, on the moving features). Panel c already says the genotype-only model sits at 0.046 ± 0.015 against a ceiling of 0.847, so the SI figure's honest reading is that node-level measurements predict a cell-level phenotype and the genotype alone, through this trunk, does not.

The 1,011 isolates. The same conditioning on natural variation needs the Muenzner 2024 proteome built beside the served Caudal 2024 transcriptomes; morphology is not measured on that panel, so the target there is the proteome from expression and the reverse. It is first in the next-build band of the expansion document and is noted here so the figure's slot exists; it is unlikely to be filled in this round.

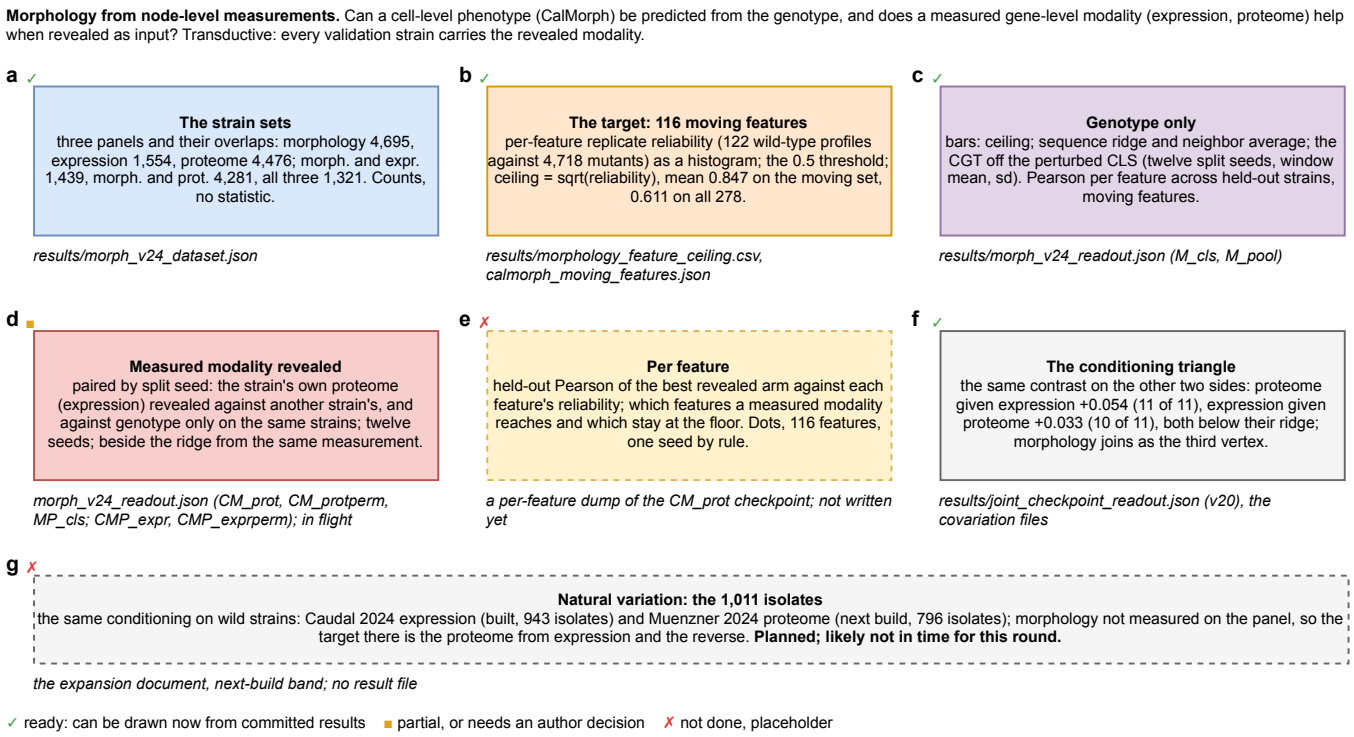


Figure 1: The morphology SI figure, sketched before its panels exist. Each box names the quantity, the statistic and the sample the panel will show, and the result file it will be drawn from; the status mark says whether that file is committed. **a**, the three single-deletion panels and their overlaps (counts). **b**, the target: per-feature replicate reliability, the 0.5 threshold that defines the 116 moving features, the ceiling. **c**, genotype only: the ceiling, the sequence baselines and the cell graph transformer read off the perturbed CLS, twelve split seeds. **d**, a measured modality revealed as input: the strain's own proteome (expression) against another strain's and against genotype only on the same strains, paired by split seed, beside the ridge from the same measurement. **e**, per feature: held-out Pearson against reliability for the best revealed arm. **f**, the conditioning triangle: the same contrast on the expression and proteome sides (v20). **g**, natural variation: the 1,011 isolates with Caudal 2024 expression and Muenzner 2024 proteome, planned.

Table 1: The morphology round's strain sets. Genotype records per store and per label, the overlaps, the split seed 0 partition, and per arm the rows left after the training script's `require_modalities` filter (the counts the runs print). A genotype is one deletion strain after deduplication and aggregation; a strain measured by two panels is one record carrying both labels.
SRC: `experiments/019-simb-multimodal/scripts/morph_v24_dataset_table.py`

store	sources	genotypes	morphology	expressi
`fig3_core`	Kemmeren 2014, Sameith 2015, Ohya 2005, Costanzo 2016 and Kuzmin 2018/2020 fitness	5,125	4,695	1,5
`fig3_morph_proteome`	Kemmeren 2014, Sameith 2015, Messner 2023, Ohya 2005	4,977	4,695	1,5
Per arm, split seed 0, after the label filter (train / validation / test rows)				
`M_cls, M_pool` on `fig3_core`		morphology: 3,757 / 469 / 469		
`CM_expr, CM_exprperm, CMP_expr, CMP_exprperm` on `fig3_core`		expression, morphology: 1,161 / 1		
`MP_cls, CM_prot, CM_protperm` on `fig3_morph_proteome`		proteome, morphology: 3,464 / 41		

Table 2: The morphology target and its references. CalMorph (Ohya 2005) gives one population-averaged profile per deletion and 122 wild-type replicates; a feature's reliability is one minus the replicate variance over the variance across mutants, its ceiling the square root, the best across-strain Pearson a perfect predictor of the signal can reach against one noisy measurement. The round predicts the moving features only. The ridge rows are the out-of-fold median Pearson per feature from the measured other modality on the strains both panels measured; the sequence rows are the best key selected on validation, four split seeds, all 278 features. SRC: `experiments/019-simb-multimodal/scripts/morph_v24_dataset_table.py`

quantity	value
deletion strains measured (Ohya 2005)	4,718
wild-type replicates	122
base CalMorph features	281
after dropping the three degenerate	278
moving features, reliability ≥ 0.5 (the target)	116
target transform	Yeo-Johnson per feature, fit on the training rows
ceiling, mean over all 278 features	0.611
ceiling, mean over the moving features (minimum)	0.847 (0.712)
ridge from the measured proteome, moving features	0.280
ridge from measured expression, moving features	0.324
sequence bilinear ridge, all 278 features	0.086
sequence neighbor average, all 278 features	0.088

Table 3: The morphology round v24: the 116 moving CalMorph features read off the perturbed CLS, on fig3_core, twelve split seeds (three for `M_pool`). Window Pearson is the mean validation Pearson per feature over epochs 1,000 to 1,199, mean ± sd over the finished runs; the paired column is the per-split difference against the arm named, mean and number of splits above zero. `M_cls` and `M_pool` are validated on every calmorph strain, the `CM` arms on the strains carrying both labels, so only the two contrasts shown are paired. The replicate ceiling of the moving set is 0.847; the sequence baselines on all 278 features (four split seeds) are 0.086 (bilinear ridge) and 0.088 (neighbor average).

SRC: `experiments/019-simb-multimodal/scripts/morph_v24_readout.py`

arm	change	finished	collapsed	window Pearson	paired difference
`M_cls`	genotype only, CLS readout, all calmorph strains	12/12	0	0.046 ± 0.015	–
`M_pool`	genotype only, CLS plus gene pool, all calmorph strains	3/3	0	0.048 ± 0.025	+0.015 (2/3) vs CLS alone
`CM_expr`	the strain's expression revealed, CLS readout (blind to it by construction)	12/12	0	0.038 ± 0.025	+0.000 (9/12) vs permuted
`CM_exprperm`	another strain's expression revealed, CLS readout (blind to it by construction)	12/12	0	0.038 ± 0.026	–
`CMP_expr`	the strain's expression revealed, pooled readout, both-label strains	4/4	0	**0.132 ± 0.111**	–
`MP_cls`	genotype only, pooled readout, proteome-and-morphology strains	2/2	0	0.068 ± 0.003	–

2 The decision ✗

Figure 3 covers single-gene deletions read out in three modalities: the transcriptome (Kemmeren 2014, 1,484 deletions), the proteome (Messner 2023, 4,476) and morphology (Ohya 2005, CalMorph, 4,718). Two framings are open and they send the next compute to different places. Framing A is a conditioning figure: one modality is measured and another is predicted from it, which is an imputation claim and needs the conditioning measurement for every strain the model is applied to (fig. 2). Framing B is a genotype-only figure: what the deletion alone predicts against each replicate ceiling and against linear and nearest-neighbor baselines (fig. 3). The choice is A, B, or neither; figs. 4 and 5 hold the supporting panels, none of which repeats a main panel, and table 4 lists what each panel lacks. The joint-training round v19 is a null on both registered tests. The conditioned round v20 is complete on 2 of 12 split seeds at 1,200 epochs, with two more below epoch 90 and eight waiting, so framing A currently rests on the ridge triangle and on two partitions (section 7). On file, the best nearest-neighbor baseline is within 0.01 of the best model score on expression and above it on the proteome and on morphology, each a maximum over keys selected on validation (fig. 2b).

SRC: `experiments/019-simb-multimodal/results/joint_checkpoint_readout.json`

Status marks. Every panel letter carries the status glyph used for section headings here: ✓ ready, the panel can be drawn now from committed results; ■ partial, or waiting on an author decision; ✗ not done, a placeholder. The marks are mockup annotations on their own draw.io layer and are switched off with `FIG3_STATUS_MARKS=0`. A *split seed* is a data partition: each seed is a different train, validation and test split, run with one initialization seed.

3 The mockups ✗

Result files are under `experiments/019-simb-multimodal/results/` unless a path is given. Each caption gives, per panel, what is plotted, the statistic and its scoring rule, the sample size and the status; the table under each figure gives the files and keys.

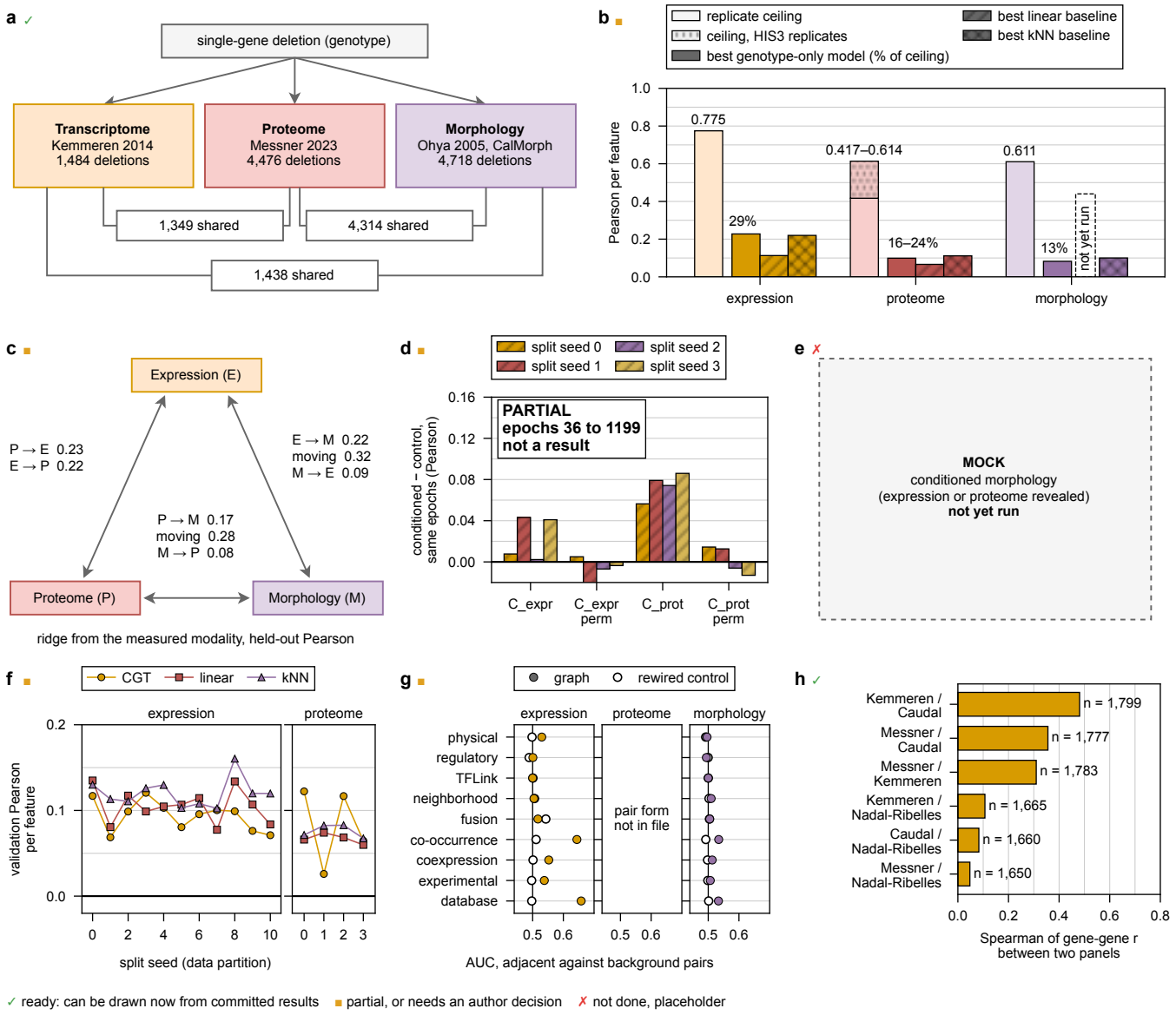
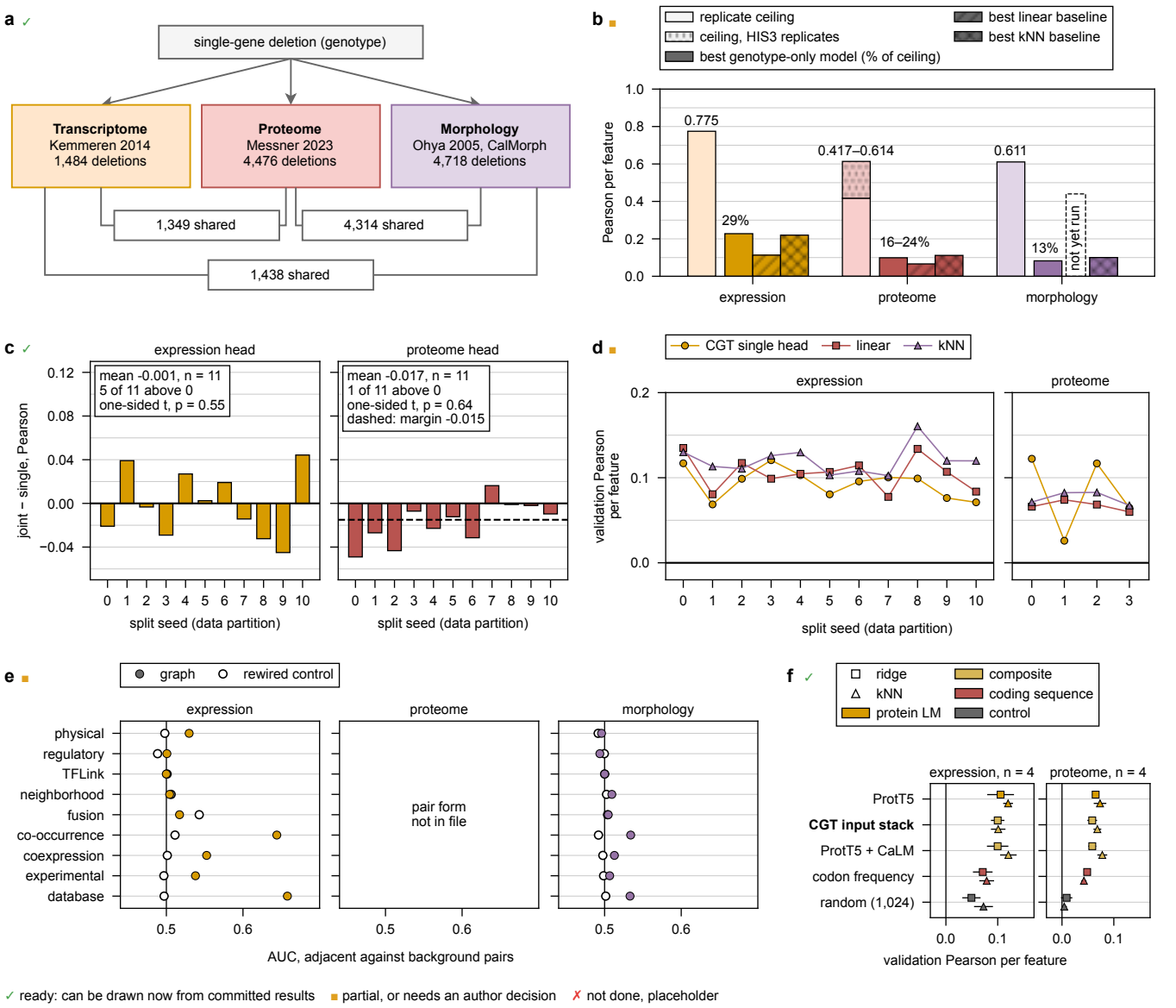


Figure 2: Ridge from one measured modality reaches a held-out Pearson of 0.17 to 0.23 on all features of another, and the conditioned model (d) is complete on 2 of 12 split seeds. **a**, The three single-deletion panels and their pairwise overlaps. Counts of distinct deletion strains; no statistic. Ready. **b**, Replicate ceiling (pale; dotted, the HIS3-replicate proteome estimator), best genotype-only model (solid; percent of ceiling), best linear baseline (diagonal hatch) and best nearest-neighbor baseline (cross hatch), as Pearson per feature across held-out strains. Model: expression and morphology are the rolling maximum of one best run over 1,353 and 183 runs, the proteome is the mean over 4 split seeds. Baselines: mean validation score over 4 split seeds of the best key (morphology, 1 split seed); no linear baseline exists for morphology. Every non-ceiling bar is a maximum over runs or keys selected on the score it reports, so it is biased upward, and the bars sit on different splits. Partial. **c**, Out-of-fold ridge between measured modalities on the deletions both panels measured (1,349, 4,314 and 1,438 strains; 5 folds): median held-out Pearson per feature; *moving* is the 116 morphology features with replicate reliability of at least 0.5. The proteome and expression side is a mean per feature from a review file. Partial. **d**, Round v20, conditioned arm minus unconditioned control: difference of the mean validation Pearson over the latest 20 epochs of each conditioned run and the same epochs of its control; 16 runs on 4 of 12 split seeds, of which split seeds 0 and 1 are complete at epoch 1,199 and split seeds 2 and 3 were at epochs 36 to 88 when read. Not a result. Partial. **e**, Conditioned morphology. Not run. Not done. **f**, Linear and nearest-neighbor baselines and the v19 single-head model per split seed (expression 11, proteome 4). Baselines: one validation Pearson per split seed, best embedding selected on that score (biased upward). Model: mean validation Pearson over the registered epoch window (unselected). A split seed is one train, validation and test partition with one initialization seed; baselines and model share the seed number and not the store, so their validation strains differ. Partial. **g**, Per graph, the AUC that two deletions adjacent on the graph give more similar phenotypes than a background pair (filled), against a degree-preserving rewired graph (open); 60,000 pairs per point, 1,482 expression and 4,718 morphology strains; no proteome strand is on file. Partial. **h**, Spearman between two panels' gene-by-gene correlation matrices, with the number of shared genes; Messner is the proteome, the others mRNA. One value per pair, no selection. Ready.

Sources for fig. 2

a	`fig3_overlap_census.json` (`kemmeren_unique`, `ohya_unique`); `fig3_proteome_build_census.json`; `n_shared_strains` of the two covariation files
b	`expression_ceiling_replicate.json`, `proteome_ceiling_replicate.json`, `morphology_noise_ceiling.json`, `round_leaderboards.csv`; `baselines_embedding_study.json` (ridge 0.113, 0.066); `graph_retrieval_baseline.json` (nearest neighbor 0.220, 0.112, STRING experimental graph key); `knn_embedding_probe.json` (0.100)
c	`proteome_morphology_covariation.json` (and its `context` block), `expression_morphology_covariation.json`
d	`joint_checkpoint_readout.json`, `v20_partial.runs`
f	`expression_baselines_split/`, `baselines_split_fig3_proteome/`; `joint_checkpoint_readout.json`, `v19.runs`
g	`graph_prior_probe.json`, `pair_form`, one step
h	`experiments/028-knockout-expression/results/proteome_expression_covariation.json`, `gene_covariation`



✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder

Figure 3: From the deletion alone the model reaches 13 to 29 % of the replicate ceiling, and the best nearest-neighbor baseline is within 0.01 of it on expression and above it on the proteome and morphology. **a**, **b**, As fig. 2a and b. Ready; partial. **c**, Round v19, joint arm minus single-head arm per split seed: difference of the mean validation Pearson per feature over the registered window (expression epochs 1,000 to 1,200, proteome 200 to 400), an unselected statistic. 11 of 12 split seeds, each a different train, validation and test partition with one initialization seed; 1,099 to 1,108 training and 119 to 126 validation strains on the four seeds on file. Expression: mean −0.001, 5 of 11 above zero, one-sided t $p = 0.55$. Proteome: mean −0.017, 1 of 11, $p = 0.64$ against the margin −0.015 (dashed). Neither registered test rejects. Ready; split seed 11 was unfinished. **d**, As fig. 2f at half width, same caveat. Partial. **e**, As fig. 2g at wide width. Partial. **f**, How much the gene representation matters to the two baselines: validation Pearson per feature, mean and s.d. over 4 split seeds, of the ridge baseline (squares) and the nearest-neighbor baseline (triangles), with rank and neighbor count selected on validation inside each seed. Five of 28 representations by rule, each the top of its class on the mean of the four validation means: best single language model (ProtT5), the model's input composite (bold), best other composite, codon frequency, best random control. The bold row is the study's `model_stack`, the concatenation `fudt_upstream`, `calm`, `prot_T5_all`, `fudt_downstream` that the v19 configuration inherits as `node_embeddings`. Ready.

Sources for fig. 3	
c	joint_checkpoint_readout.json, v19.tests (H1a_expression_superiority, H1b_proteome_noninferiority); strain counts from split_gene_overlap_audit.json
f	baselines_embedding_study.json; the composite is defined in experiments/019-simb-multimodal/scripts/expression_baselines_split.py and matched to conf/cgt_expr_v13_split.yaml

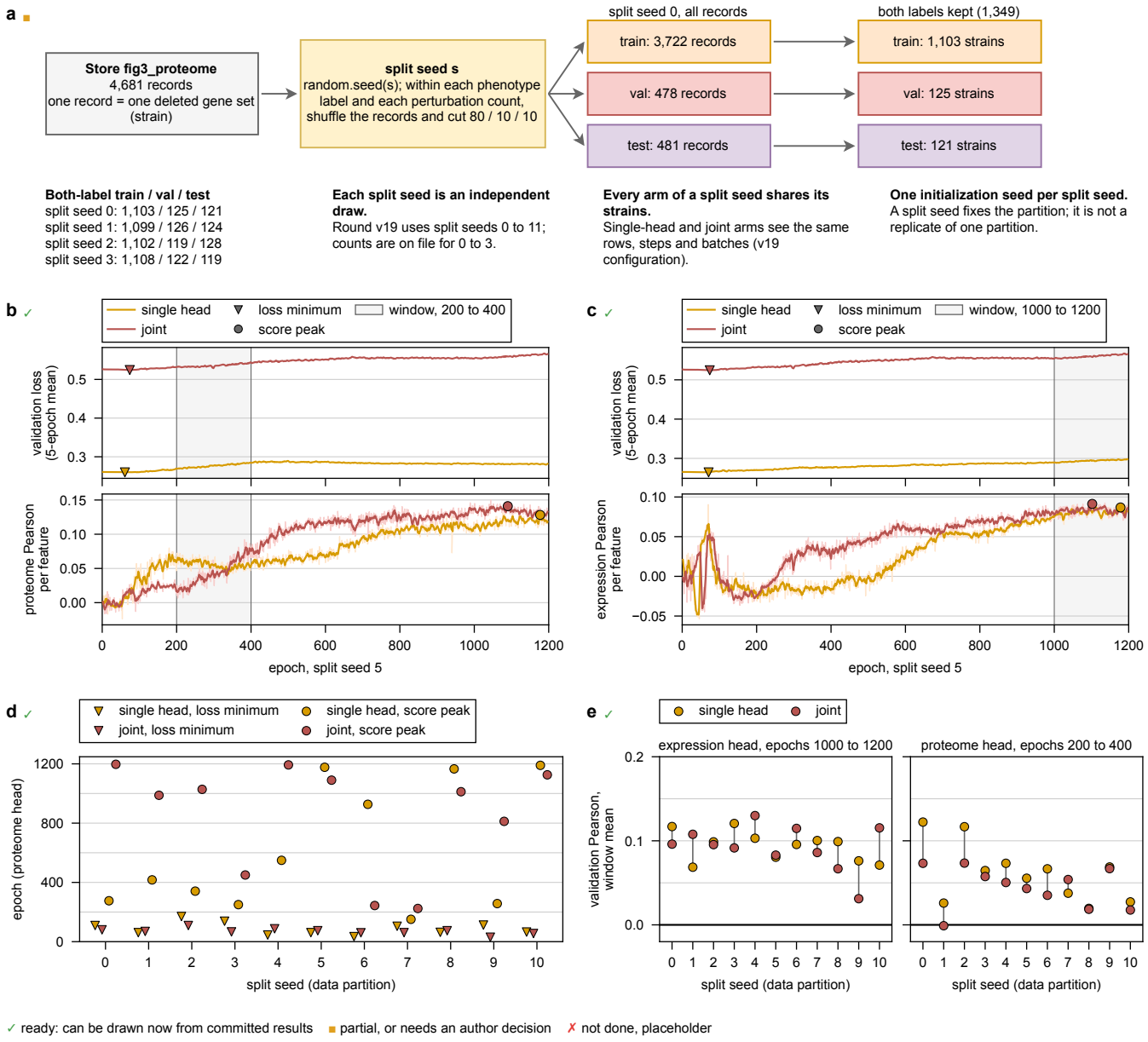


Figure 4: In round v19 the validation loss reaches its minimum before epoch 200 on every run, and the proteome score peaks later on every run. a, How a split seed becomes the strains of a run, drawn from the code. The store's 4,681 records (one record is one deleted gene set) are shuffled within each phenotype label and each perturbation count and cut 80, 10, 10 under the split seed; each side then keeps the records carrying both labels, 1,349 in all, so the both-label fractions are near and not exactly 80, 10, 10. Counts for split seeds 0 to 3. No statistic. Partial: seeds 4 to 11 are not on file. **b**, Proteome head, split seed 5, one run per arm: the 5-epoch centered mean of the validation loss (top) and the validation Pearson per feature (bottom; pale raw, dark its 5-epoch mean) against epoch. Triangles, loss minimum; circles, maximum of the 5-epoch mean, an upward-biased order statistic; shaded, the registered window. The joint loss sums two heads, so the two loss levels are not comparable. Split seed 5 is chosen by rule: its joint-minus-single proteome difference (−0.012) is the median over the 11 complete split seeds. About 1,100 training and 125 validation strains. Ready. **c**, The same for the expression head of that split seed, window epochs 1,000 to 1,200. Ready. **d**, Proteome head, all 11 complete split seeds, 22 runs: epoch of the loss minimum (triangles) and of the score peak (circles). Ready. **e**, The window scores behind fig. 3c: mean validation Pearson over the registered window, single-head and joint arm, 11 split seeds, 33 runs. Unselected. Ready.

Sources for fig. 4

a	`torchcell/datamodules/cell.py` (`_compute_and_save_index`); `experiments/019-simb-multimodal/scripts/train_cgt_multitask.py` (`require_modalities`); counts from `split_gene_overlap_audit.json`
b, c	`joint_checkpoint_curves.csv` (41,129 rows: run, arm, split, epoch, loss, its 5-epoch mean, each head's Pearson), written by `experiments/019-simb-multimodal/scripts/joint_checkpoint_readout.py`; marker epochs from `joint_checkpoint_readout.json`
d, e	`joint_checkpoint_readout.json`, `v19.runs`

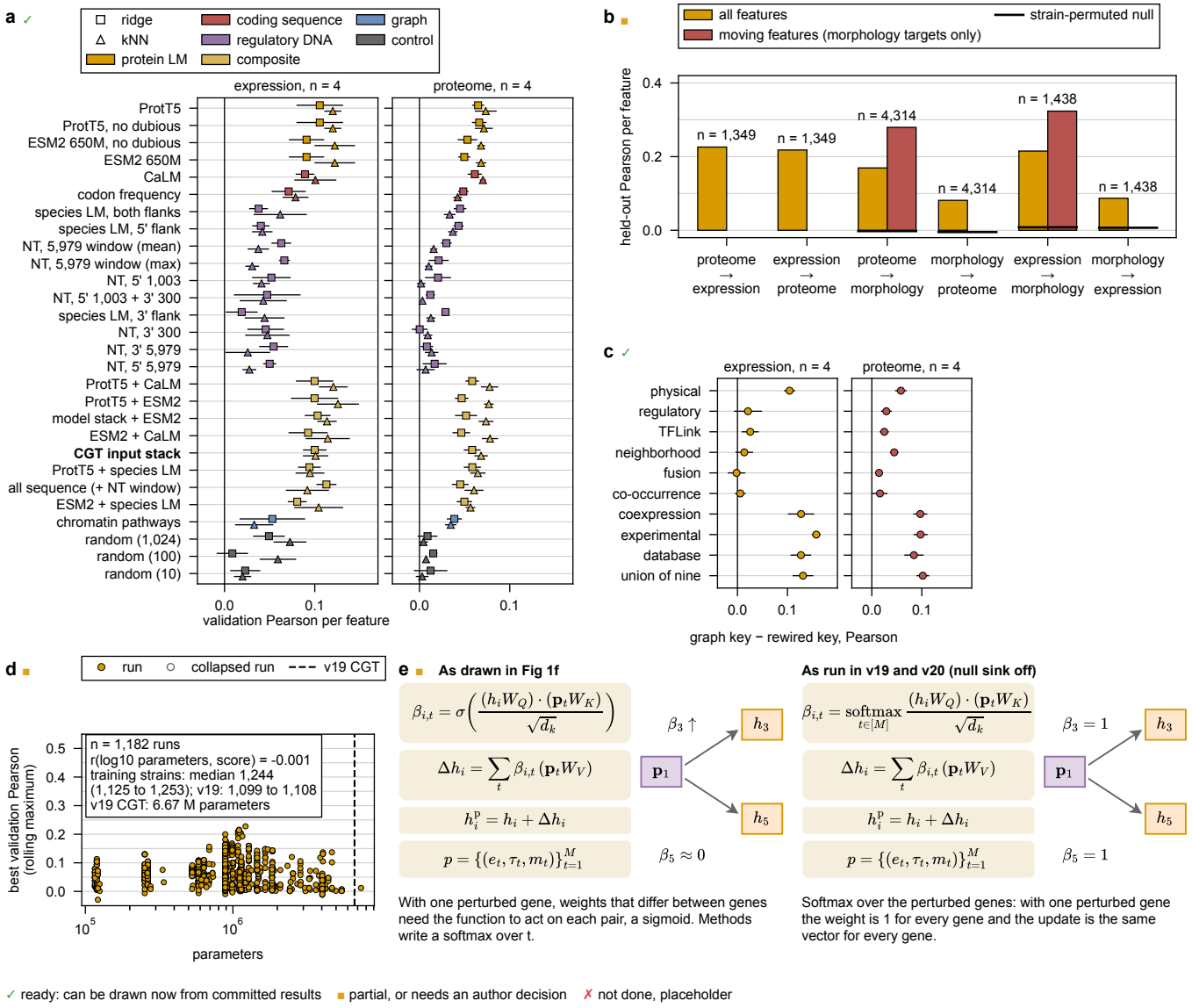


Figure 5: The model's input composite scores below ProtT5 alone in both baselines on both modalities, and the best score of a run does not track its parameter count. a, All 28 gene representations of the baseline study, by family: validation Pearson per feature, mean and s.d. over 4 split seeds, ridge (squares) and nearest neighbor (triangles), expression (1,244 to 1,253 training strains) and proteome (3,581); rank and neighbor count selected on validation inside each seed. Bold, the model's input composite. Ready. **b**, The six directed pairs of fig. 2c as bars with the strain-permuted null (black line, on file for four pairs) and the shared-strain count. Partial. **c**, The graph as a retrieval key: validation Pearson of a neighbor average keyed on each graph's adjacency row minus the same with the rewired graph, mean and s.d. over 4 split seeds, expression and proteome. Neighbor count selected on validation. Ready. **d**, Parameter count against the rolling maximum of the validation Pearson, one point per expression-strand run (1,182 runs with both values; open, collapsed runs). The rolling maximum is biased upward and the runs differ in epochs run. The Pearson between $\log_{10}$ parameters and score is -0.001. Dashed, the v19 model at 6.67 million parameters for about 1,100 training strains, near 6,000 parameters per strain; its score is a window mean and is not placed on this axis. Partial. **e**, The perturbation operator in the notation of manuscript Fig. 1f; no data. Left, as drawn there. Right, as the code computes it and as the Methods write it: a softmax over the perturbed genes, which with one perturbed gene gives weight 1 to every gene. Partial, an author decision.

Sources for fig. 5

a	`baselines_embedding_study.json`
b	as fig. 2c, plus `triangle` of `joint_checkpoint_readout.json` for one strain count
c	`graph_retrieval_baseline.json`, rounds `v13` and `v14`, `summary.paired`
d	`round_leaderboards.csv` (`total_param_count`, `primary_roll_max`); `total_param_count` of `v19.runs` in `joint_checkpoint_readout.json`; `split_gene_overlap_audit.json`
e	`torchcell/models/equivariant_cell_graph_transformer.py` (`EquivariantPerturbationTransform`); `paper/nature-biotech/sections/methods.tex`

4 What each panel still needs ✗

Table 4: What each panel still needs before it is real. ✓ ready: every drawn value is in a committed results file and the panel can be drawn now. ■ partial: a run is unfinished, scoring rules or splits are mixed, or an author decision is pending. ✗ not done: nothing behind it has been run.

panel		what remains
A and B, a	✓	The expression and morphology overlap reads 1,438 in `expression_morphology_covariation.json` and 1,440 in `fig3_overlap_census.json`; reconcile.
A and B, b	■	No linear baseline for morphology is on file. The bars mix scoring rules and splits. Rescore model and baselines under one rule on one set of split seeds, and choose one proteome ceiling estimator.
A, c; SI-2, b	■	The proteome and expression side is a mean per feature from a review file, with no permuted null and no moving subset on file; recompute it with the estimator of the other two sides, on a common strain set.
A, d	■	2 of 12 split seeds complete at 1,200 epochs, 2 below epoch 90, 8 waiting on `cabbi`. Needs all split seeds, the registered window, and the ridge of panel c as the reference to beat.
A, e	✗	No store joins the proteome and morphology, and the observed-label channel takes gene-level values, so morphology can be a target and not yet an input.
A, f; B, d	■	Baselines and model share a seed number and not a store, and the baselines are single selected validation scores while the model is a window mean. Rerun the baselines on the both-label store. The proteome baselines cover split seeds 0 to 3 only.
A, g; B, e	■	The pair-form probe has no proteome strand; add it to `graph_prior_probe.py`.
A, h	✓	Sameith is in `gene_covariation_all.json` and not drawn.
B, c; SI-1, e	✓	Split seed 11 was unfinished at readout; redraw when it is read. The result is a null.
B, f; SI-2, a	✓	Four split seeds; the baselines ran on the full expression and proteome stores.
SI-1, a	■	Both-label counts are on file for split seeds 0 to 3 of 12. That every arm sees the same batches is stated by the v19 configuration and was not re-verified here.
SI-1, b, c	✓	One split seed by rule; the other ten are in `joint_checkpoint_curves.csv`.
SI-1, d	✓	The score peak is a rolling maximum, an upward-biased order statistic.
SI-2, c	✓	Four split seeds per strand.
SI-2, d	■	The expression document's "145 runs, −0.129" is produced by no committed script and no subset of `round_leaderboards.csv` tried here reproduces it; the panel draws all 1,182 scored expression runs, $r = -0.001$. Scores are rolling maxima over unequal budgets. A v19 point needs a v19 score under the same rule.
SI-2, e	■	Author decision: Fig. 1f draws gene-dependent weights with one perturbed gene, and the Methods and the code compute a softmax, which gives weight 1. Either the figure or the operator changes. The null sink, which restores a sigmoid weight, has one trained comparison stopped at 186 to 189 epochs.

5 What the audit of 2026-10-04 established ✗

Ten independent read-only reviews of the expression, proteome and morphology work were run on 2026-10-04, one scope each, and are filed under `review/2026-10-04/`. Where two reviews reached the same point from different files it is stated once. Every number is measured unless it is marked as a hypothesis.

No trained model on file beats the best baseline on matched strains. On the test strains of the model's own partitions, read at the best-validation checkpoint, the graph transformer scores 0.110 and 0.153 on expression (split seeds 0 and 2) against 0.165 and 0.174 for a nearest-neighbor average keyed on the interaction graphs, and 0.067 and 0.082 on the proteome against 0.117 and 0.111. The model minus the best validation-selected key is negative in six of six comparisons, by 0.02 to 0.06. This rests on two split seeds per modality, because test predictions were saved for six of forty checkpoints; validation orders the same way on all four split seeds. Morphology has no matched comparison at all. A second review reads a different set of checkpoints (v17 reference arm, three seeds per split) at a model edge of about +0.017 over the matched sequence baselines, most of it from split seed 0, so the sign against sequence baselines is unsettled and the sign against the graph key is not. SRC: `review/2026-10-04/04_baselines_and_ceilings.md,07_data_splits_metrics.md`

Headline numbers that do not hold. The incumbent's 0.196±0.022 is the mean of eight different arms of one round, all at seed 0 on split seed 0, scored by a rolling maximum; same-seed reruns agree within 0.001, and no configuration has been replicated at a long budget. Split seed 0 is a favorable draw: its validation score sits 1.72 standard deviations above the mean of twelve baseline draws, and every long run used it. The claim that the model clears the bilinear baseline by 0.093 compared a rolling maximum against a baseline scored on a different permutation of the strains. The rank-64 pair term's advantage per unit of compute compared one run in each of two projects and fails at matched budgets. The statistic "145 runs, correlation of log parameters with score −0.129" has no generating script and is not reproduced by `round_leaderboards.csv`, which gives −0.001 over 1,182 runs confounded with budget. SRC: `review/2026-10-04/01_expression_pre_simb.md,02_expression_post_simb.md,06_training_dynamics.md,09_prior_reviews_and_literature.md`

The abstract's two correlations are retracted. The manuscript and the SIMB 2026 abstract give $r = 0.543$ for knockout expression and $r = 0.619$ for single-knockout morphology. Both are wrong and neither is a result. The author's

account (2026-10-05) is that they came from earlier runs with the wrong splits and the wrong data processing. The record agrees on every point it can check: no committed script regenerates either number (`plan.simb-2026-multimodal-cgt.2026.07.21`, correction to WS10); the morphology value was computed on a superseded build of the Ohya store (`torchcell.datasets.scerevisiae.ohya2005`); the expression store behind the value held 1,450 records where the current one holds 1,484 deletions; and the best values measured since on held-out strains are 0.238 and 0.082 per feature, both rolling maxima. The two numbers were removed from the abstract, the Results filler and the outline board on 2026-10-05 and are not to be quoted as prior performance. SRC: `review/2026-10-04/09_prior_reviews_and_literature.md`, `notes-tex/019-simb-multimodal/sections/8-directions.tex`

What has a measured effect. Budget, within a single run: 0.147 at epoch 500, 0.202 at 2,000, 0.230 at 4,000. The partition: between-partition standard deviation 0.020 to 0.030 against 0.010 to 0.020 within, larger than any arm effect. The masked objective: removing it costs 0.012 over twelve pairs with a confidence interval that excludes zero, negative on all four partitions; rounds v19 and v20 run without it. Embedding content against width-matched random vectors: +0.068. Interaction graphs as a neighbor key against their rewired controls: +0.150 on expression and +0.115 on the proteome, four of four split seeds. Among sequence representations ProtT5 alone is at or above every composite, and the four-part stack the model consumes adds −0.03 to +0.003 over it. No architecture change has met the campaign's own bar of +0.02 on three of four partitions with the test sign agreeing. SRC: `review/2026-10-04/01_expression_pre_simb.md`,`02_expression_post_simb.md`,`04_baselines_and_ceilings.md`

What the code does at one deletion. The v19 model has 6,681,548 parameters, of which 5.69 million, 85 percent, are the first linear layer that projects the 3,328 dimensional concatenated embedding; the encoder is 0.59 million and the perturbation operator 0.10 million. The encoder runs once on the wild-type graph, so its output is the same for every strain. The operator is a softmax over the perturbed genes; with one deleted gene its weight is identically 1 and every gene receives the same vector, so its query and key parameters (16,380) receive zero gradient. The Methods describe this softmax; Fig. 1f draws a weight that differs between genes, which this operator cannot produce at one deletion. The operator's attention dropout acts on that single weight and drops whole heads of the deletion signal at random. Nothing marks which gene was deleted: the model predicts +0.036 for the deleted gene's own transcript against a measured −2.42, and a predictor that writes only that one entry scores 0.0175 per-feature Pearson, the size of the edge in dispute. The null sink, the sigmoid form that makes the weight depend on the querying gene, has four attempts on file: two with the gate nearly closed at initialization, stopped at 78 to 189 epochs; four seed pairs with the gate open at 300 epochs, +0.0024 (sd 0.0090); and one run with the gate open at 4,184 epochs (W&B `fqk5mbr3`), 0.200 against 0.210 for its paired reference by rolling maximum, one seed on split seed 0. It delivers at most nine gene-dependent scalars, one per head, and has not been replicated at budget. SRC: `review/2026-10-04/05_architecture_and_operator.md`,`07_data_splits_metrics.md`,`01_expression_pre_simb.md`

What the training curves show. The runs are neither converged nor memorizing. Train-side Pearson is still rising at epoch 1,200 and flattens below 0.78 after 10,000 epochs. The validation-loss minimum near epoch 70 to 90 sits at the end of a plateau where predictions are still near-constant, so it marks the launch of the head, not the onset of overfitting, and the loss-minimum checkpoint is close to a mean predictor. Predictions are over-dispersed for their correlation (spread ratio about twice the Pearson), which is why the normalized error exceeds 1; a term that pushes the prediction spread toward the target's would raise that error further. Wall time does not depend on model size: an epoch costs the same at 0.68 and 1.45 million parameters, a fifth of it is the eval-mode train pass, and the encoder is recomputed every step although its output never changes. With the encoder output cached a CPU step falls from 42 s to 2.6 s, and to 0.23 s on 512 reporter genes. SRC: `review/2026-10-04/06_training_dynamics.md`,`05_architecture_and_operator.md`

Defects found. Three bear on numbers already reported. The prediction-dump routine calls the model without the observed-label input, so every dumped validation and test prediction from the masked-objective lineage comes from a different function than the logged metric. The v16 expression baselines were taken from the wrong store's partitions. The per-feature Pearson drops near-constant columns per model, so two arms can be scored on different feature sets. One bore on running jobs: the conditioned arms were launched without persistent data-loader workers, the setting whose absence killed four v15 runs; the pending v20 tasks were resubmitted with it (job 2423297) and tasks 0 and 1 of job 2423190 run as launched. SRC: `review/2026-10-04/05_architecture_and_operator.md`,`07_data_splits_metrics.md`,`03_proteome_and_joint.md`

What new data can and cannot do inside the window. The local SPELL archive holds 200 studies with genetic perturbations covering 1,175 genes, of which 478 are not among the 1,484 Kemmeren deletions: 216 are promoter knockdowns of essential genes, 146 come from Hughes 2000, and 14 from Hu 2007. On the ridge learning curve 245 to 460 more strains project to about +0.004 to +0.008 (hypothesis, one split), below the between-partition spread. The natural isolates need a reference record, per-allele sequence embeddings that no builder supports, and a loader fix, five to eight days before a first number. The two amino-acid screens do not agree with each other because the Cooper screen does not reproduce its own duplicate strains. No relationship between fitness and the expression or proteome profile has been measured. SRC: `review/2026-10-04/08_cross_modal_and_new_data.md`

The conditioned round. The conditioning step is wired as intended: validation and test are conditioned, standardization is fit on train, the revealed head leaves the loss and the metrics. Its comparison against the v19 single-head

arms is not like for like (head slot, loader setting, and an encoder input of values against zeros), so the primary contrast is each conditioned arm against its permuted control, and the bar is the ridge from genotype plus the measured modality refit per partition (0.236 on expression and 0.214 on the proteome on one split), not the unconditioned arm.
SRC: review/2026-10-04/03_proteome_and_joint.md

6 The plan to the deadline ✗

Fewer than twenty days remain and compute is not the constraint. The pooled Delta allocations hold about 28,800 unused GPU hours (`bfjt` 11,162, `bflt` 9,904, `bgcg` 4,622, `bbub` 3,145), and IGB adds two `cabbi` cards and up to four A40s on `gpu`. A full-size run of 1,200 epochs costs about 8.5 GPU hours when four share an A40 (about 34 hours for a pack of four, v19 timing), so twelve split seeds of one arm cost about 100 GPU hours. Everything below fits in under a quarter of the allocation (table 5). The plan therefore runs five tracks at once. Nothing waits on another track's result unless it needs that result as input, and the dates are the days on which the content of the figure is decided, not days on which work stops. Nothing is submitted to Delta until the author says so; every job is an individual submission with no dependency chain.

Table 5: The plan uses under a quarter of the unused Delta allocation. Run counts are arms × twelve split seeds. Cost per run is the v19 timing on an IGB A40 with four runs per card, scaled linearly with epochs; the cost on a Delta card is a hypothesis until the first canary reports. SRC: thissection;Deltabalancesread2026-10-04

track	what	runs	epochs	GPU hours
B, wave 1	seven single-change arms and the reference	96	1,200	820
B, long budget	reference and two arms	36	4,000	1,020
B, wave 2	combinations and screen winners, up to eight arms	96	1,200	820
B, wave 2 long	up to three arms	36	4,000	1,020
C	retrain reference and best arm on each enlarged store, three stores	72	1,200	610
D	morphology on its own 4,718 strains, four arms	48	1,200	1,600
E	conditioned round, morphology sides added	48	1,200	410
total		432		about 6,300

The morphology row assumes four times the per-run cost of the 1,349-strain store because the training set is about four times larger (hypothesis, untimed).

Track A, days 0 to 4, CPU only: the matched benchmark and the rescoring. Four pieces. First, the linear and nearest-neighbor baselines for every gene representation on exactly the strains rounds v19 and v20 use, twelve split seeds, both labels (IGB jobs 2423298 to 2423302, running). Second, the *rescoring harness*: one script that re-scores every model already trained under a single rule, so that any two numbers in the figure are comparable. It dumps predictions through the function the logged metric uses, blanks the deleted gene's own entry, scores a feature set fixed in advance by replicate reliability, reads window means in place of rolling maxima, uses test strains, and bootstraps over strains. Third, a learning-curve harness that tells the candidate limits apart: a ridge and a bilinear model on 137 to 1,100 training strains over five splits, and the same on planted labels at the measured noise level. Fourth, two gates that earlier reviews asked for and no one ran: perturbation-derived embeddings of the deleted gene (its genetic-interaction, chemogenomic, morphology or proteome profile) through the existing baselines, and measured fitness as an input to the genotype ridge with a shuffled control. **Decision, day 4:** the matched table is drawn with ceilings, and it is the first row of Figure 3 whatever the other tracks return.

Track B, from day 1: the model, at full size and in a fast screen at once. The single-change arms the audit agrees on are launched at full size on twelve split seeds as soon as each passes a canary, without waiting for a screen: the masked objective restored; the sigmoid-gated operator with a term in the reporter and the perturbation and an indicator of the deleted gene; the deleted gene's graph neighborhood as input; the projection cut from 5.69 million parameters to a frozen low-dimensional one; ProtT5 alone against the stack; operator attention dropout off; the perturbed CLS as a global readout, ported from the interaction branch. The reference and two arms also run at 4,000 epochs, since budget is the one lever with a measured effect inside a run. In parallel the *small-model harness* screens the combinations: the encoder runs once on the wild-type graph, its output is cached to disk, and only the operator and the heads train on top of it, so one card screens dozens of variants a day at 300 epochs on three split seeds, scored by the mean over epochs 200 to 300. Hypothesis (untested): the cached model runs at least ten times faster per epoch on a GPU; the measured figure is 16 times on a CPU. **Decision, day 8:** wave 2 takes the combinations the screen ranks highest and every wave-1 arm that beats its paired reference on at least nine of twelve split seeds.

Track C, from day 1: more deletion-keyed expression. The loaders are written now, in parallel with the model work, since they cost no GPU: the SPELL deletion and knockdown sets in order of new genes (the 216 promoter knockdowns of essential genes, Hughes 2000 with 146, then the remainder of the 478), each through the modular dataset

path with its provenance record, and the natural isolates against their own reference strain. The ridge learning curve projects +0.004 to +0.008 from the SPELL strains on one split (hypothesis); the transformer's response to more strains has not been measured, and retraining the reference and the best arm on each enlarged store measures it. The isolates need a reference record and per-allele sequence embeddings, five to eight days of work before a first number, and their store build depends on the dataset-build machine. **Decision, day 12:** a store enters the figure if its retrained arms are finished on twelve split seeds.

Track D, from day 3: morphology on its own strains. Morphology has no matched comparison and has only been trained on the 1,349-strain overlap. Its own 4,718 strains are trained with the perturbed CLS readout against the gene-token readout, with the same matched baselines as Track A.

Track E: the conditioned round. v20 runs to completion on the two `cabbi` cards. The store joining the proteome with morphology is built so that the two strongest sides of the triangle, expression to morphology and proteome to morphology, can be run as conditioned arms. **Decision, about day 10:** with at least six split seeds, a conditioned arm enters the figure if it exceeds its permuted control by 0.02 and reaches the per-partition ridge; otherwise it is reported as a model that uses a second modality less well than a linear map.

Days 12 to 19. Day 12, the content of the figure is fixed from what has finished on twelve split seeds. Days 12 to 16, at most three configurations rerun with a sealed test set. Days 16 to 19, freeze, read the sealed test set once, draw the figure, write.

Out of the window. A pooled single-cell screen, and the distribution-matching loss term, which the audit predicts would raise the error.

Decisions that are the author's. Whether Figure 3 may be a benchmark in which a graph neighbor average ties or beats the transformer, if that is what day 12 shows. Whether Fig. 1f or the Methods changes, since they name different operators. When Delta submissions start, and on which account.

7 Status on the evening of 2026-10-05 ✗

What has finished, what is running, and what each framing rests on tonight. The mockups of section 3 were redrawn from the result files as they stand at 22:30; the only panel whose data changed since the morning is fig. 2d.

Round v19 is complete and null on both registered tests. All twelve split seeds finished training on IGB; the readout on file reads eleven, split seed 11 having been unfinished when it was taken. Expression at the registered window (epochs 1,000 to 1,200), joint minus single head: mean −0.001, standard deviation across partitions 0.030, 5 of 11 above zero, one-sided t test $p = 0.55$. Proteome at epochs 200 to 400: mean −0.017, 1 of 11 above zero, $p = 0.64$ against the non-inferiority margin of −0.015. Joint training on identical both-label rows buys nothing on either head at this budget. fig. 3c and fig. 4e draw these eleven; the twelfth is added when the readout is rerun. SRC: `joint_checkpoint_readout.json,v19.tests`

Round v20 is complete on two of twelve partitions. Tasks 0 and 1 of IGB job 2423190 reached epoch 1,199 after 31 hours each on `cabbi`; tasks 2 and 3 were at epochs 36 to 88 at 22:25 and tasks 4 to 11 wait under a two-at-a-time throttle. On the two complete partitions, each statistic the mean validation Pearson per feature over the latest 20 epochs of the conditioned run against the same epochs of the unconditioned v19 arm:

head	given	split seed	conditioned	unconditioned	permuted control
proteome	expression	0	0.168	0.112	0.126
proteome	expression	1	0.095	0.016	0.029
expression	proteome	0	0.125	0.117	0.122
expression	proteome	1	0.126	0.083	0.003

The proteome head conditioned on the strain's own expression sits +0.056 and +0.079 above the unconditioned arm, and the permuted-expression control gains only +0.014 and +0.013, so the gain uses the strain's own measurement and not the presence of a second input. The expression head conditioned on the proteome gains +0.008 and +0.043, and on split seed 0 its permuted control gains as much (+0.005), so there is no separation there yet. The bar the plan set is the per-partition ridge from genotype plus the measured modality, 0.236 on expression and 0.214 on the proteome on one split; both conditioned arms are below it on both partitions. Two partitions are two draws from a distribution whose between-partition spread is 0.02 to 0.03 (section 5); this is a reading, not a result. W&B runs `7tiaq7pe`, `fg63qx22` (conditioned proteome), `3o61ceay`, `izzszopm` (its permuted control), `7a7tc6xb`, `qvmr4wyl` (conditioned expression), `zxlow2i4`, `u6p781n5`, project `torchcell_019_prot_v20`. SRC: `joint_checkpoint_readout.json,v20_partial.runs`

Track A: the matched baselines have run and are unread. IGB job 2423302, the linear and nearest-neighbor baselines on exactly the strains of the both-label store at twelve split seeds, finished on 2026-10-04 and has not been read. Until it is, fig. 2f and fig. 3d keep the caveat that baselines and model share a seed number and not a store.

Depth and width do not set the epoch time. A three-hour window on the four GilaHyper cards (RTX 6000 Ada, 44.4 GiB) was spent timing the model at reduced size rather than training it, since convergence cannot be read in three hours. Four cells at zero data-loader workers, four to eight runs sharing a card: the v19 model on the 3,328-d stack (6.68 million parameters), ProtT5 alone at the same trunk (1.45 million), ProtT5 at three layers (1.15 million), and ProtT5 at width 54 with two layers (0.74 million). Card memory was 8 to 10 GB per run at every shape, so eight runs on a card ran out of memory (three of eight), six sat at the limit, and four is the ceiling; the memory is the activations over 6,600 gene tokens at batch 32, not the parameters. Steady-state epoch time at four per card was 78 s for the full model and 80 to 81 s for every ProtT5 shape. The same held with the cards fed: with three persistent workers per run the epoch fell to 35 to 40 s at four per card and did not move with depth. Delta's own v11 round logged 81 to 134 s per epoch at one run per A40 with zero workers. So at both worker settings the trunk is not where the time goes. The runs are in W&B project `torchcell_019_small_bench` (GilaHyper jobs 3267, 3271 and 3273) and `torchcell_019_grid_bench` (job 3274). SRC: `notes/experiments.019-simb-multimodal.scripts.gh_small_model_bench.md`

cell (four runs per card)	workers per run	seconds per epoch per run	card memory
3,328-d stack, width 90, six layers	0	78	36 GB
ProtT5, width 90, six layers	0	80 to 81	36 GB
ProtT5, width 90, three layers (six per card)	0	80	at the limit
ProtT5, width 54, two layers (eight per card)	0	81	out of memory
ProtT5, width 90, six layers	3	35 to 40	38 GB
ProtT5, width 90, six layers	1 (eight CPUs)	58 to 64	38 GB
ProtT5, width 90, two layers	3	37 to 40	33 GB
ProtT5, width 90, three layers	3	38 to 41	37 GB
ProtT5, width 90, four layers	3	39 to 43	36 GB
ProtT5, width 90, six layers	3	39 to 42	37 GB

The last four rows are the depth-by-operator grid (job 3274) at epoch 9, each row the range over its four operators; the operators sit within 3 s of each other at every depth. The persistent-worker setting is the one lever found: the Delta rule of zero workers dates from July, when workers were re-spawned every epoch and re-imported torch from Lustre; persistent workers pay that import once. Hypothesis (untested): the time scales with batch size times gene count, through the node tensors, the per-gene readout and loss, the metric passes and the eval-mode passes, and one place it can go is the perturbation operator, which the code runs as a Python loop over the strains in a batch (`torchcell/models/equivariant_cell_graph_transformer.py`), one cross-attention over all genes per strain; a one-step profile would settle it. The cached-encoder speedup of section 5 was measured on a CPU and does not transfer to a GPU on this evidence.

Decision taken: the small trunk is ProtT5 alone at width 90 and six layers. Since depth and width buy no time and no packing, the v21 configuration (`conf/cgt_expr_v21_small.yaml`) changes one thing against v19, the input: ProtT5 alone, 1.45 million parameters against 6.68, with every other setting inherited. Hypothesis (untested until v21 reads out): the smaller trunk reaches the v19 single-head level on these rows.

Round v21 is on Delta. Track B, wave 1, submitted at 21:58 on `bfjt-delta-gpu` at source 9ca207582: nine arms on twelve split seeds, 108 runs, as 27 independent single-A40 jobs of four runs each (jobs 22690625 to 22690651), one persistent worker per run on eight CPUs, 1,200 epochs, a two-day limit, no dependency between jobs. A canary of four reference copies at three epochs with a test pass (job 22690624) runs beside them; if it fails, the round is canceled. The queue was faster than the scheduler's estimate: `sbatch -test-only` placed the first pack at 16:08 the next day, and fifteen packs were running by 22:50. The canary completed in 1 h 42 min with the test pass logged on all four runs, about 87 minutes of it the store import from Lustre; pack 0 under the round's settings read 49 to 59 s per epoch per run at epoch 8, so a pack finishes in about 22 to 24 h against the 48 h limit (a projection from epoch 8). That projection was wrong: it read the training progress bar, and the logged epoch time was 177 s (section 8). The arms, each one change against the reference S_ref (the v21 configuration as is): S_mask, the masked objective restored; S_sink, the sigmoid-gated operator with the gate open; S_prop2, the deletion propagated two hops along the graphs; S_nodrop, operator attention dropout off; S_stack, the four-part embedding stack in place of ProtT5 (the size control); S_prot, the single-head proteome model on this trunk; S_basis64, the rank-64 response basis; S_hadam, the elementwise product at the operator. Scoring is pre-registered in the configuration header: the mean of the validation Pearson per feature over epochs 1,000 to 1,200 (the proteome arm over 200 to 400); an arm is carried to wave 2 if it beats S_ref on at least nine of twelve split seeds; a run whose prediction spread ratio is below 0.05 at epoch 200 is relaunched with a new initialization seed. The store is v19's: fig3_proteome restricted to the 1,349 genotypes carrying both labels, synced to Delta tonight. SRC: `experiments/019-simb-multimodal/scripts/delta_expr_v21_small.slurm`, `conf/cgt_expr_v21_small.yaml`, `scripts/gh_expr_008_arm.sh`

The operator question has three candidates under test. fig. 5e still waits on the author's choice between Fig. 1f and the Methods. Three operators that give each gene its own response at one deletion are in v21 at twelve paired seeds each (S_sink, S_hadam, S_basis64) and in the GilaHyper grid tonight on split seed 0 at four depths. The grid's operator read, validation Pearson and its spread at about epoch 200 by operator within each depth, follows. Sixteen runs on one split seed cannot decide the operator; they size what v21 decides.

The grid at epochs 119 to 132: a partial read at the launch stage. Job 3274 was cut from two hours to 96 minutes to fit the window and ended by time limit with all sixteen runs alive, at epoch 119 (six layers) to 132 (two layers). Every number below is from a stage where the heads have barely launched: the validation Pearson per feature sits at 0.00 to 0.04 against 0.09 to 0.13 for the same arm at 1,200 epochs in v19, and the prediction spread ratio is 0.05 to 0.27 against the launch gate of 0.05. One split seed, one initialization. Not a result. The four operator columns are the validation Pearson per feature, mean of the last 20 epochs:

layers	S_ref	S_sink	S_hadam	S_basis64	s per epoch
2	0.010	0.017	**0.020**	0.007	41.9
3	0.013	0.018	**0.033**	0.017	43.5
4	0.004	0.010	**0.029**	0.012	44.4
6	0.034	0.032	0.039	**0.039**	45.8

Seconds per epoch are the mean over the four operators of each run's mean after epoch 5, including the eval-mode train pass every tenth epoch; the four operators are within 1 s of each other at every depth. Depth costs about one second per layer per epoch, a tenth of the epoch between two and six layers, and that is the whole throughput case for a shallower trunk. Two readings, each on one seed at one tenth of the budget, and so hypotheses to be read against v21: the Hadamard operator is above the softmax reference at all four depths (by 0.005 to 0.025) while its eval-mode train Pearson is the lowest at every depth (0.15 to 0.22 against 0.17 to 0.25), so it fits the training strains less and the held-out strains more at this stage; and the six-layer trunk, which has run the fewest epochs, is above the shallower ones on every operator at this stage, with the lowest train fit. The rolling maximum reads 0.046 to 0.063 on every run and

separates nothing. W&B project `torchcell_019_grid_bench`; the depth 3 Hadamard run is `ydyxxub2`, its reference `gfzwmdq8`. SRC: `scratchpad/grid_bench_read.py`, `job3274tasklogs`

What this leaves for the two framings. Framing A rests on two complete partitions of v20 and the ridge triangle; the conditioned proteome arm clears its permuted control by 0.04 to 0.07 on both and is below the ridge bar on both. Framing B is unchanged: v19 is null, the matched baselines are on disk unread, and no trained model on file beats the nearest-neighbor key on matched strains. The first number that can move either framing is the v21 readout on 2026-10-07 and 2026-10-08, and the next partitions of v20 as `cabbi` releases them.

8 The night of 2026-10-05: speed, the Delta round, and the fast round v22 ✗

Correction: v21 on Delta cannot reach its window as launched. Section 7 projected 22 to 24 hours per pack from the training progress bar, 49 to 59 s. The logged epoch time is 177 s on every run: the reference arm was at epoch 69 after 3.3 hours of training. At that pace 1,200 epochs take 59 hours against the 48 hour limit, and a pack stops near epoch 945, short of the registered window of epochs 1,000 to 1,200. The cause was read from forty stack dumps of one run over six minutes: in all forty the main thread sat in the eval-mode train pass, spawning a data-loader worker. That pass rebuilt its loader every tenth epoch, and on Delta a spawned worker re-imports the software stack from Lustre, more than twelve minutes each time. It follows from launching the round with one loader worker; with zero the pass would have been slow but would not have respawned anything. The epoch times are `perf/epoch_seconds` of W&B project `torchcell_019_expr_v21`; the dumps were taken on Delta job 22690625. SRC: `notes/experiments.019-simb-multimodal.scripts.v22_readout.md`

Where an epoch went. A profile of the reference run on the GilaHyper cards, then stack samples of live runs, located five costs, none of them the model's own arithmetic (table 6). Producing a sample cost 50 ms (a database read, a JSON parse, the reconstruction of an experiment record with thousands of phenotype values) and was repeated for every sample every epoch. Decoding each strain's target row from the batch ran as a Python loop over the strains with about ten points per strain where the CPU waits on the GPU, 59 to 65 percent of a training process's time once several processes shared a card. The perturbation operator ran one strain at a time. The epoch metrics ranked a 1,100 by 6,127 matrix on the CPU three times an epoch. And all six encoder layers built a full gene-by-gene attention matrix that only the one graph-regularized layer reads. Each was replaced by a form that computes the same thing, held equal to the original by a test: the operator and the decode on batched tensors, the ranks on the GPU, the fused attention kernel on the unregularized layers, and the splits held in memory after one pass. After the changes the stack samples show the process waiting on the GPU, on the operator's feed-forward block over every gene of every strain; what is left is the architecture's own work. SRC: `notes/experiments.019-simb-multimodal.scripts.gh_profile_cgt_expr.md`

Table 6: The same 1,200-epoch run is four times faster on Delta and the batch-128 form is twice as fast again. Seconds per epoch per run, from `perf/epoch_seconds` over at least 20 epochs, with the runs sharing a card as stated. *Before* is the code v21 was launched on.

machine and card	run	runs per card	s per epoch, before	s per epoch, after
Delta, A40	reference, batch 32	4	177	45
GilaHyper, RTX 6000 Ada	reference, batch 32	3	29 to 32	24
GilaHyper, RTX 6000 Ada	batch 128	3	19 to 20	11.4

A batch-128 run holds 12.2 GB of card memory against 8.5 GB at batch 32, so three fit on a card and a fourth runs out of memory. Processes sharing a card take turns on it, so a card of mixed arms runs every arm at the pace of the slowest; each card below holds one arm.

The state of v21 on Delta. The queue's start estimate for a resubmission was a day out, so one pack was replaced inside its own running job at 03:58: the four reference runs of split seeds 0 to 3 were stopped at epoch 70 and restarted from scratch on the new code with zero loader workers and the splits in memory. They log 38 to 41 s per epoch, 100 s on every tenth, about 45 s on average, which is 15 hours for 1,200 epochs. The other 26 packs were left as launched. Replacing them the same way is prepared and was run without effect as a check; it needs the author's decision, as does the alternative of canceling and resubmitting through the queue. The four restarted runs are W&B `iuqwl2p8`, `tdnhpkid`, `fcq0xkke` and `0jajrluv`. SRC: `experiments/019-simb-multimodal/scripts/delta_v21_inplace_swap.sh`

The rising validation loss on expression is a scale problem. Earlier sections record that the validation loss bottoms out early and rises while the per-feature Pearson keeps climbing, and that at its Pearson peak no run beats the per-gene mean on squared error. A predictor with correlation r to its target minimizes squared error when the ratio of its spread to the target's equals r. On the six saved validation prediction sets, one scalar was fit on half the validation strains (with a per-gene intercept) and scored on the other half (table 7). The expression predictions are 1.4 to 2.2 times too spread for their correlation; shrinking them by the fitted scalar brings all three checkpoints below the per-gene mean, to what the correlation allows (for split seed 0, $1.036 \times (1 - 0.225^2) = 0.983$ against a measured 0.984). The proteome head has the opposite sign at its best-validation checkpoint and is under-dispersed. So the ordering the metric reads is intact and the magnitudes are overfit; calibration can be repaired after training without changing the Pearson. The same arithmetic states how little is explained: r^2 is 2 to 5 percent of per-gene variance across held-out strains.

Round v22: what a faster batch costs. Every arm is the v21 reference with one change, on the new code, 1,200 epochs unless stated, scored as v21 by the mean validation Pearson per feature over epochs 1,000 to 1,200 (tables 8 and 9). Two to three split seeds per arm size an effect and do not decide one.

Table 7: One scalar fit on half the validation strains brings every expression checkpoint below the per-gene mean on squared error. NMSE is the squared error per gene over that gene's variance across the scoring strains, averaged over genes; 1 is the per-gene mean of the scoring strains, and the mean estimated on the fitting half reads above 1. Each row is a best-validation checkpoint (a selected epoch) of round v13 or v14; halves swapped and averaged. The lowest NMSE of each row is bold.

checkpoint	val strains	val Pearson per feature	sd(pred) / sd(target)	fitted scalar c	NMSE, per-gene mean	NMSE, model as is	NMSE, shrunk by c
expression, V_ref_s2	155	0.144	0.317	0.46	1.030	1.052	**1.016**
expression, V_ref_s0	155	0.225	0.319	0.70	1.036	0.993	**0.984**
expression, V_concat_s0	155	0.192	0.343	0.52	1.036	1.029	**0.997**
proteome, P_concat_s0	448	0.113	0.062	1.75	1.010	**0.997**	1.002
proteome, P_ref_s0	448	0.116	0.085	1.38	1.010	**0.995**	0.997
proteome, P_ref_s2	448	0.128	0.063	1.89	1.008	0.992	**0.987**

Table 8: Batch 128 trains as far as the reference in half the time when its learning rate is left alone or warmed up, and collapses when it is scaled without warmup. A run is *collapsed* when its prediction spread ratio never reached 0.05 in 300 epochs, or reached it and then stayed below 0.01 for 50 consecutive epochs, the per-gene-mean predictor; such runs were stopped. The highest arm mean is bold; an arrow marks an arm whose complete runs are all above or all below the reference on shared split seeds.

arm	change against the reference	split seeds run	collapsed or never launched	complete at 1,200 epochs	mean val Pearson per feature, epochs 1,000 to 1,200 (complete runs)	mean paired difference vs F_ref (n shared split seeds)	s per epoch
F_ref	the v21 reference: batch 32, lr 3×10^{-4}, six layers, width 90	3	0	3	0.062	+0.000 (3)	24.0
F_b128lr1	batch 128, lr 3×10^{-4}	11	0	11	**0.082**	+0.013 (3) ↑	13.5
F_b128	batch 128, lr 6×10^{-4}	3	2	1	0.059	+0.000 (1)	10.2
F_b128wu	batch 128, lr 6×10^{-4} after a 50-epoch warmup	3	0	3	0.067	+0.005 (3)	11.4
F_b128lr4	batch 128, lr 1.2×10^{-3}	3	3	0			12.0
F_l4w180	batch 32, four layers, width 180	2	2	0			17.6
F_b128wu_hadam	F_b128wu with the Hadamard operator	3	0	3	0.036	-0.025 (3) ↓	11.2
F_b128wu_wd	F_b128wu with weight decay 0.3, 800 epochs	3	0	0			11.4
F_b128wu_drop	F_b128wu with dropout 0.3, 800 epochs	3	1	0			11.4
F_b256lr1	batch 256, lr 3×10^{-4}, 1,000 epochs	1	0	0			3.1

Table 9: Every run of v22, with its W&B link. The three ladder columns are the mean validation Pearson per feature over the 20 epochs ending at that epoch, so runs stopped early can still be compared at matched epochs. A run with no entry in the window column did not finish 1,200 epochs; the state column gives the last epoch reached in parentheses. The highest window score is bold.

			val Pearson per feature, mean of the 20 epochs ending at epoch						
arm	W&B run	split seed	200	600	1,199	val Pearson, mean over epochs 1,000 to 1,200	spread ratio, last epoch	train Pearson, last eval-mode pass	state (last epoch)
F_ref	o053grx1	0	0.004	0.007	0.061	0.055	0.378	0.561	complete
F_ref	7cmz3r5i	1	-0.017	0.026	0.071	0.071	0.349	0.522	complete
F_ref	60puupp0	2	0.095	0.033	0.070	0.059	0.413	0.513	complete
F_b128lr1	671rcqge	0	0.014	0.013	0.076	0.069	0.401	0.525	complete
F_b128lr1	jock61jk	1	-0.014	0.005	0.083	0.075	0.314	0.509	complete
F_b128lr1	rza3mz2y	2	0.115	0.032	0.085	0.081	0.444	0.535	complete
F_b128lr1	yp83dtne	4	0.043	0.027	0.090	0.085	0.387	0.501	complete
F_b128lr1	gmg3l6ji	5	0.026	0.037	0.072	0.067	0.525	0.534	complete
F_b128lr1	3r4pypib	6	0.025	0.097	0.153	**0.141**	0.364	0.519	complete
F_b128lr1	z82tcdl9	7	0.061	0.081	0.116	0.104	0.332	0.516	complete
F_b128lr1	65ctt125	8	0.108	0.060	0.077	0.076	0.384	0.518	complete
F_b128lr1	hdnhxjyc	9	0.004	0.022	0.057	0.051	0.388	0.511	complete
F_b128lr1	wbh7zn1y	10	0.055	0.058	0.098	0.094	0.327	0.513	complete
F_b128lr1	ua44739t	11	0.019	0.049	0.073	0.065	0.511	0.500	complete
F_b128	l885wzyv	0	0.028	0.005			0.038	0.062	collapsed at 212 (625)
F_b128	zz9sbrr3	1	0.006	-0.006			0.000	0.019	collapsed at 658 (930)
F_b128	z5fxcrnx	2	0.103	0.026	0.065	0.059	0.435	0.524	complete
F_b128wu	49zyp3vw	0	0.007	0.019	0.087	0.081	0.447	0.592	complete
F_b128wu	yokxkvjc	1	-0.019	0.019	0.085	0.071	0.358	0.527	complete
F_b128wu	i4n88yyy	2	0.103	0.042	0.061	0.049	0.355	0.461	complete
F_b128lr4	rbn6guq7	0	0.003				0.000	0.021	never launched (442)
F_b128lr4	jzx4kljl	1	0.002				0.000	0.007	never launched (454)
F_b128lr4	a9mz22oq	2	0.001				0.000	0.018	never launched (446)
F_l4w180	pbqtgnpe	0	-0.004				0.000	0.008	collapsed at 215 (312)
F_l4w180	ntn060cz	1	-0.007				0.000	-0.153	collapsed at 119 (305)
F_b128wu_hadam	56m6vbt8	0	0.029	0.005	0.067	0.048	0.374	0.509	complete
F_b128wu_hadam	j3l6huyu	1	-0.013	-0.007	0.019	0.013	0.270	0.410	complete
F_b128wu_hadam	wcp7wryw	2	0.099	0.047	0.044	0.048	0.411	0.499	complete
F_b128wu_wd	18gw3p46	0	0.011	0.009			0.077	0.152	complete
F_b128wu_wd	k38jsllm	1	-0.020	0.009			0.206	0.320	complete
F_b128wu_wd	t9avzqjw	2	0.089	0.039			0.348	0.332	complete
F_b128wu_drop	w6sy4dlf	0	0.022	-0.004			0.178	0.282	complete
F_b128wu_drop	w08c6cda	1	-0.013	0.000			0.000	-0.000	collapsed at 213 (799)
F_b128wu_drop	jo6tr18b	2	0.110	0.058			0.139	0.259	complete
F_b256lr1	qqphe2y4	0	0.046	-0.003			0.322	0.385	complete

Batch 128 needs its learning rate left alone or warmed up. At four times the reference rate no run launched (3 of 3 split seeds): the prediction spread stayed at zero from the first epoch. At twice the rate without warmup two of three runs launched and then returned to the per-gene-mean predictor, at epochs 212 and 658. At the unscaled rate, and at twice the rate behind a 50-epoch linear warmup, all six runs completed; their window scores are 0.069, 0.075, 0.081, 0.085, 0.067, 0.141, 0.104, 0.076, 0.051, 0.094, 0.065 (mean 0.082) and 0.081, 0.071, 0.049 (mean 0.067) on split seeds 0, 1 and 2. The batch-32 reference on the same code reads 0.055, 0.071, 0.059 (mean 0.062). Against it the unscaled arm is +0.013 on average, 3 of 3 split seeds above (per split seed +0.014, +0.003, +0.022), and the warmup arm is +0.005 on average, 1 of 3 split seeds above (per split seed +0.026, −0.000, −0.009). Three split seeds resolve only a difference whose sign all three share, which table 8 marks with an arrow; the partition moves the score more than either arm does. What the batch buys is time: 11.4 s per epoch against 24, three runs to a card in both cases. SRC: `experiments/019-simb-multimodal/results/v22_readout.json`

A collapse is a return to the per-gene mean, and the validation loss rewards it. In the collapsed runs the gradient norm stays between 0.01 and 0.05 and the clip at 10 never engages, so nothing diverges; the prediction spread falls to zero within about 30 epochs and the validation loss returns to 0.2506, the loss of the per-gene-mean predictor. The runs that keep training sit at a validation loss of 0.254 to 0.257, above it. On held-out strains the loss prefers the collapsed predictor to the trained one, which is the scale problem of table 7 seen from the other side. The collapsed runs are `l885wzyv`, `zz9sbrr3`, `pbqtgnpe` and `ntn060cz`; a run that kept training is `o053grx1` (table 9 links each). SRC: `notes/experiments.019-simb-multimodal.scripts.gh_expr_v22_fast.md`

Four layers at width 180 did not train at the reference rate. Both split seeds launched and collapsed, at epochs 215 and 119, and were stopped near epoch 310. No wider trunk has yet been run with a warmup or a lower rate, so this is a statement about one setting.

The Hadamard operator is behind the softmax operator at this budget. On the warmup arm, with the same split seeds, the Hadamard operator of v21's S_hadam reads 0.048, 0.013, 0.048 (mean 0.036) at the window, against the softmax operator −0.031 on average, 0 of 3 split seeds above (per split seed −0.033, −0.058, −0.002). Its heads launch later and its curves are still the steepest at epoch 1,199, so the read is of this budget; in the two-hour grid of section 7 it led at epoch 120, which shows how little an early read predicts. v21 carries the same contrast at batch 32 on twelve split seeds.

Every curve dips and is still climbing at 1,200 epochs. On each surviving run the validation Pearson rises early (to 0.10 to 0.12 near epoch 200 on split seed 2), falls through epochs 400 to 800 while the train Pearson climbs past 0.4, and rises again; between epochs 1,000 and 1,199 the 20-epoch mean still gains about 0.02 (the ladder columns of table 9). The registered window therefore reads a budget and not a plateau, as the audit of section 5 found for the longer runs.

Two regularizers at 800 epochs. On the warmup arm, weight decay raised from 10^{-8} to 0.3 and dropout raised from 0.1 to 0.3 were each run for 800 epochs and read against the warmup arm at matched epochs. At epoch 400, in the dip, weight decay is +0.001 on average, 3 of 3 split seeds above (per split seed +0.001, +0.001, +0.000), and dropout is +0.015 on average, 3 of 3 split seeds above (per split seed +0.002, +0.008, +0.034). At epoch 799 weight decay is +0.010 on average, 3 of 3 split seeds above (per split seed +0.020, +0.006, +0.004), and dropout is −0.006 on average, 1 of 3 split seeds above (per split seed −0.026, −0.015, +0.024). Runs collapsed: 0 of 3 under weight decay and 1 of 3 under dropout. The 800-epoch runs stop before the registered window, so these are matched-epoch reads and not window scores.

Two further speed levers, timed for 40 epochs each. With three batch-128 runs on a card the new code runs a plain epoch in 8.8 to 9.5 s and an epoch that carries the eval-mode train pass in 21.5 to 22.5 s. Narrowing the operator's feed-forward block from four times the width to one takes the plain epoch to 8.0 to 8.5 s, about a tenth, and frees 3.4 GB of the card; whether the narrower block scores the same is not measured. A single batch-256 run alone on a card takes 3.0 s per plain epoch and 6.0 s with the eval pass, in 19.6 GB, so two fit on a card and a 1,200-epoch run takes about an hour. One batch-256 run at the unscaled rate was then trained for 1,000 epochs on split seed 0: at epoch 799 it is −0.006 on average, 0 of 1 split seeds above (per split seed −0.006) against the batch-128 unscaled arm and +0.001 on average, 1 of 1 split seeds above (per split seed +0.001) against the reference, one split seed each. The timing runs are in W&B project `torchcell_019_profile` (GilaHyper job 3332). SRC: `notes/experiments.019-simb-multimodal.scripts.gh_profile_cgt_expr.md`

What follows for the next rounds. New rounds should run on the new code with the splits in memory, zero loader workers on Delta, and one arm per card. Batch 128 at the unscaled rate is the candidate default if its difference from the reference above is acceptable to the author: it halves the time per epoch and has not collapsed in three runs. Any arm that raises the learning rate needs a warmup. Run time is to be projected from the logged epoch seconds over at least 20 epochs, never from the progress bar. The decisions that are the author's: whether the other 26 packs of v21 are replaced in place, resubmitted, or left to stop near epoch 740 to 820 (the 15:33 read: 230 s per epoch, not the 177

s of the first eleven hours); and whether the registered window should move later, given that every curve is still rising at 1,200.

9 What the expression predictions look like ✗

The scores above are per-gene Pearson summaries. Figure 6 shows the predictions themselves for the best expression checkpoint on file, the v13 reference on split seed 0 (validation Pearson per gene 0.223), beside the bilinear ridge (0.134) and the nearest-neighbor average (0.130) on ProtT5, refit on the same 155 validation strains and 6,127 graph genes with the cells selected on validation.

The pooled correlation is mostly the gene means. The 0.45 of panel a is not a model score. The per-gene train mean alone, repeated for every strain, correlates 0.40 with the measurements across all pairs, because genes differ in their mean log2 ratio (the gene means have sd 0.09 against a pooled sd of 0.22 and carry 16.7 percent of the pooled variance), and the model's predictions correlate 0.77 with that mean profile. With the gene means removed from both sides (panel b) the pooled correlation is 0.28 and the mean per-gene Pearson is 0.22.

Direction, not distance. Within a gene the model's deviations have a third of the measured spread (median sd ratio 0.32, panel e) and a regression slope of 0.09 on the measured value. The nearest-neighbor average has the full spread (1.08) at a lower correlation (0.17). The ratio that minimizes squared error is the Pearson itself, 0.22, so the model is 1.4 times too spread for squared error and three times too narrow against the measurements; both statements describe the same 0.32. Per-gene Pearson rises with how much a gene moves (panel d): from 0.1 on the quietest genes to about 0.4 on the most variable for the model, 0.1 to 0.2 for both baselines.

No predictor reaches the large responses. The strongest validation strain (panel f, the YER068W deletion) spans measured log2 ratios from -2 to $+2.5$ across genes; neither predictor leaves ± 0.5. A median strain (panel g) is the gene-mean profile with a small offset. The model's best gene (panel h, YKL145W, Pearson 0.62) has its strains in the right order over a predicted range of 0.0 to 0.15 against a measured -1 to 0.5.

panel	files and keys
all	`prediction_scatter.json`: `predictors` (per-gene mean and median Pearson, pooled raw and centered Pearson, median spread ratio, centered slope), `pooled_decomposition`, `examples`; predictions from `val-predictions/compute-0-2-2397312\_3fcfb279\ldots.json`; baseline cells from `expression\_baselines\_split/seed0.json`

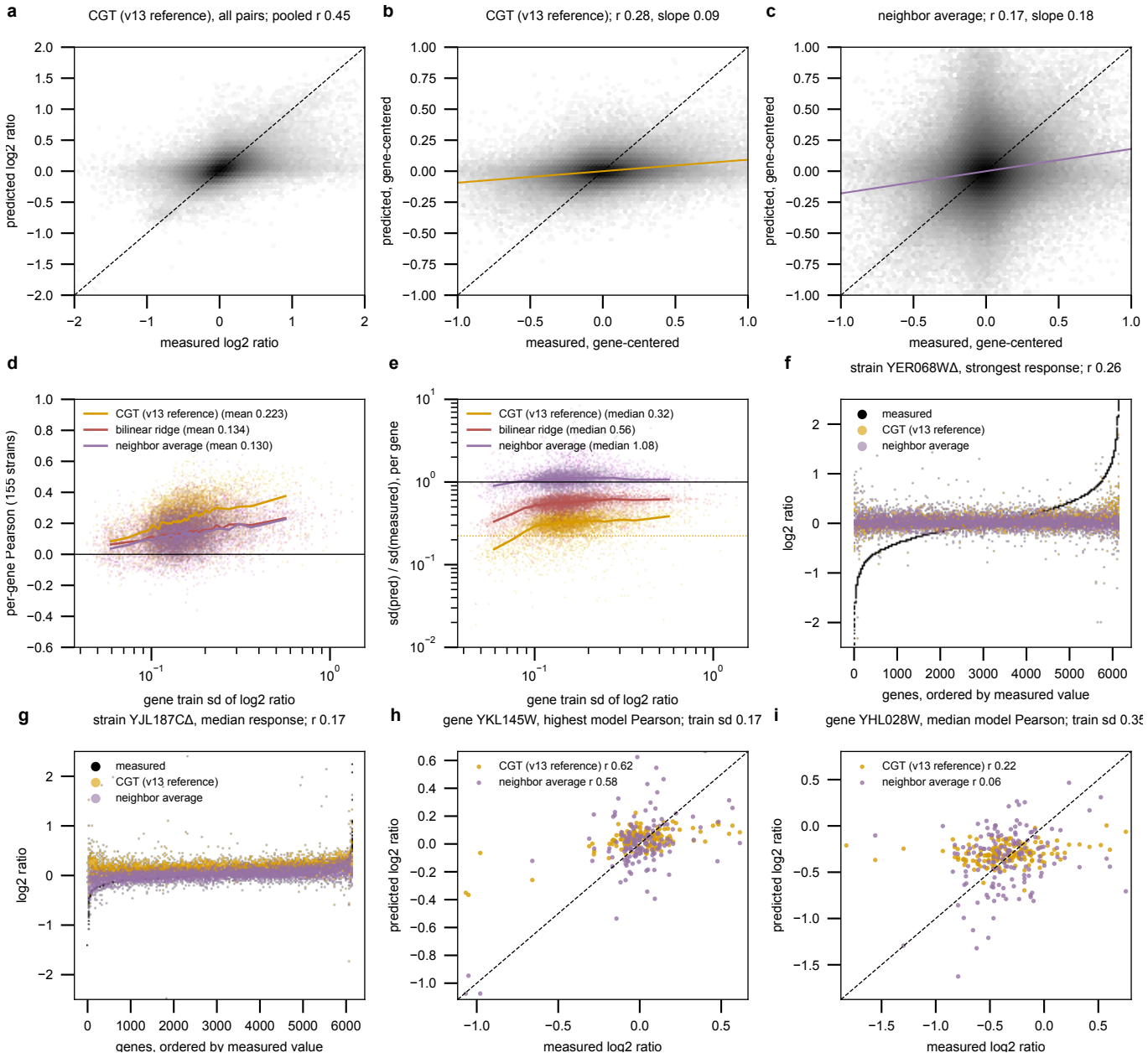


Figure 6: The v13 reference predicts the direction a gene moves across strains and not the distance; the large responses are missing from every predictor. Validation strains of split seed 0 on fig3_core (155 strains, 6,127 graph genes). CGT: v13 reference checkpoint, run wq8y8nd5; ridge and neighbor average: B2 and B3 on ProtT5, cells selected on validation. **a**, Predicted against measured log2 ratio, every strain-gene pair (hexbin, log count); pooled Pearson 0.45. **b**, The same with each gene's train mean removed from both axes; pooled Pearson 0.28, regression slope 0.09 (orange). **c**, The neighbor average on the axes of b; 0.17, slope 0.18. **d**, Per-gene Pearson over the 155 strains against the gene's train sd, with the running median per predictor (40 bins); means 0.223, 0.134, 0.130. **e**, Per-gene sd(predicted) / sd(measured) on the same axis; the solid line is 1, the dotted line 0.22 is the model's mean Pearson; medians 0.32, 0.56, 1.08. **f**, **g**, One strain each, genes ordered by the measured value: the strongest-response strain (YER068W deletion, sd over genes 0.56, per-strain Pearson 0.26) and a median one (YJL187C deletion, 0.17, 0.17). **h**, **i**, One gene each over the 155 strains: the gene with the highest model Pearson (YKL145W, 0.62; neighbor average 0.58) and the gene at the model's median (YHL028W, 0.22; 0.06). Dashed lines are the identity. One checkpoint, one partition; no uncertainty is shown.

10 2026-10-07, 02:00: v20 at six partitions, v21 relaunched, and the collapse census ✗

v20 is complete on four partitions and live on two. Tasks 2 and 3 of the conditioned round finished at epoch 1,199 on `cabbi` after 21 h each; tasks 4 and 5 are at epochs 447 to 498 and tasks 6 to 11 wait under the two-task throttle, about 21 h per pair. table 10 is the readout at six partitions. The proteome head conditioned on the strain's own expression sits above the unconditioned arm of the same partition at the same epochs on 6 of 6 (by +0.018 to +0.097, mean +0.055), and above its permuted-expression control on 6 of 6 (by +0.034 to +0.101, mean +0.054): the gain uses the strain's own measurement. The expression head conditioned on the proteome is above the unconditioned arm on 5 of 6 (mean +0.031; split seed 2 read −0.049 at epoch 1,135 and −0.005 at 1,199) and above its permuted control on 6 of 6 (mean +0.045, split seed 0 by 0.003). Both conditioned heads remain below the pre-registered per-partition ridge from the measured other modality (0.236 expression, 0.214 proteome on one split): the model uses the other modality, and less efficiently than a linear map from it. Splits 4 and 5 are partial.

The 26 untouched v21 packs were relaunched in place. The author's decision was the in-place swap. From 18:38 to 18:51 `delta_v21_inplace_swap.sh` stopped the slow runs of every untouched pack and relaunched four fresh runs each inside the same jobs, 104 runs on the code of pack 0. Startup took 65 minutes with 104 runs importing from `/work/hdd` and materializing their splits from one LMDB over Lustre at once (pack 0 alone took 5 minutes); training began near 19:45 at a median 38 s per plain epoch and about 44 s with the eval-mode train pass, and at 01:26 the runs were at epochs 353 to 481. The median run ends near 10:40 on 2026-10-07 and the slowest near 15:00, against a limit near 22:00. Pack 0 finished its two surviving runs at 1,199 and was canceled by hand: its watcher's `pgrep` matched its own command line and would never have fired. The relaunched packs end with `scancel` after the pack script returns.
SRC: `notes/user.Mjvolk3.torchcell.tasks.weekly.2026.41.multimodal-phenotype-retrospective.md`

Collapse is a one-epoch event, frequent, arm-dependent, and rarely reversed. table 11 counts every dead stretch of the two small-trunk rounds. The spread ratio falls from 0.1 to 0.3 to below 0.01 between consecutive epochs, and the validation loss is equal or lower afterwards, so the per-gene-mean predictor is a stable basin rather than a divergence. At the 01:50 read, of 54 stretches of 50 epochs or more, 10 later returned to a spread of 0.05, and of the 29 of 100 or more, 3 (the table is regenerated on the finished rounds; section 11 gives the final counts). Gradient clipping (threshold 10) never fired in any run; the per-epoch gradient norm at the collapse epoch is a median 1.14 times the mean of the 50 epochs before (quartiles 0.96 to 1.48, 5 of 54 at 2 or more), so a gradient spike is NOT the general trigger, which corrects the reading of three cases on the evening of 2026-10-06. Onsets concentrate between epochs 150 and 350, right after launch. By arm, on the relaunched v21 round at epochs 491 to 663: Hadamard 8 of 12 collapsed, no dropout 5, null sink 5, stack 4, rank-64 basis 3, reference 1 of 8, and the mask schedule, the propagation arm and the proteome 0 of 12 each. Fourteen collapsed or never-launched runs of the pre-swap round were stopped by hand on 2026-10-06; the relaunched round's collapsed runs are still running. Hypothesis (untested): the Hadamard and no-dropout arms collapse because their per-gene product leaves the perturbation path without a bounded gate; the null sink's 5 of 12 argues against the gate being sufficient.

The twelve-seed baselines on the both-label store. IGB job 2423302 ran the ridge and nearest-neighbor baselines on the strains carrying both labels, the store v21 and v22 train on, over all twelve split seeds (table 12). On expression the nearest-neighbor average on ProtT5 reaches 0.110±0.017 validation and 0.090±0.027 test; on the proteome 0.065±0.013 and 0.040 ± 0.024. Every v22 window score (0.062 to 0.075) and every v21 trailing score read so far sit below the expression baselines on this store. The same job's expression-label run on the whole fig3_proteome store had overwritten the committed protein-abundance seeds 0 to 3 in `baselines_split_fig3_proteome_full/`; those were restored from the repository, the expression files moved to `baselines_split_fig3_proteome_expression_log2_ratio_full/`, and the writer now keys a non-default label into the directory name.

Table 10: PARTIAL: the conditioned round at the epoch each run had reached when this table was generated, of a 1,200-epoch budget. *conditioned* is the mean of the predicted head's validation `pearson_per_feature` over the run's latest 20 epochs; *control* is the v19 single-head arm of the same partition over the same epochs; the last column is that control at its registered window. `C_expr` predicts expression with the strain's proteome revealed and `C_prot` the reverse; the `perm` arms reveal another strain's measurement. Nothing here is a result until the runs finish.

SRC: `experiments/019-simb-multimodal/scripts/joint_checkpoint_readout.py`

arm	split	predicted	epoch	conditioned	control	diff	control, window
C_expr	0	expression	1200	0.125	0.117	+0.008	0.117
C_expr	1	expression	1200	0.127	0.083	+0.043	0.069
C_expr	2	expression	1200	0.098	0.102	−0.004	0.099
C_expr	3	expression	1200	0.157	0.118	+0.038	0.121
C_expr	4	expression	1200	0.152	0.114	+0.038	0.103
C_expr	5	expression	1200	0.103	0.081	+0.022	0.080
C_expr	6	expression	1199	0.127	0.098	+0.029	0.096
C_expr	7	expression	1199	0.143	0.104	+0.039	0.100
C_expr	8	expression	1199	0.144	0.106	+0.038	0.099
C_expr	9	expression	1199	0.114	0.076	+0.039	0.076
C_expr	10	expression	1199	0.159	0.089	+0.070	0.071
C_expr	11	expression	1199	0.120	–	–	–
C_exprperm	0	expression	1200	0.122	0.117	+0.005	0.117
C_exprperm	1	expression	1200	0.003	0.083	−0.080	0.069
C_exprperm	2	expression	1200	0.090	0.102	−0.012	0.099
C_exprperm	3	expression	1200	0.129	0.118	+0.011	0.121
C_exprperm	4	expression	1200	0.085	0.114	−0.030	0.103
C_exprperm	5	expression	1200	0.072	0.081	−0.010	0.080
C_exprperm	6	expression	1199	0.078	0.098	−0.020	0.096
C_exprperm	7	expression	1199	0.092	0.104	−0.012	0.100
C_exprperm	8	expression	1199	0.098	0.106	−0.008	0.099
C_exprperm	9	expression	1199	0.083	0.076	+0.007	0.076
C_exprperm	10	expression	1199	0.078	0.089	−0.011	0.071
C_exprperm	11	expression	1199	0.075	–	–	–
C_prot	0	proteome	1200	0.168	0.112	+0.056	0.122
C_prot	1	proteome	1200	0.095	0.016	+0.079	0.026
C_prot	2	proteome	1200	0.139	0.112	+0.027	0.117
C_prot	3	proteome	1200	0.125	0.071	+0.054	0.065
C_prot	4	proteome	1200	0.113	0.090	+0.022	0.073
C_prot	5	proteome	1200	0.180	0.120	+0.060	0.055
C_prot	6	proteome	1199	0.087	0.073	+0.014	0.067
C_prot	7	proteome	1199	0.096	0.040	+0.056	0.038
C_prot	8	proteome	1199	0.128	0.037	+0.091	0.020
C_prot	9	proteome	1199	0.135	0.066	+0.069	0.069
C_prot	10	proteome	1199	0.111	0.050	+0.061	0.027
C_prot	11	proteome	1199	0.134	–	–	0.028
C_protperm	0	proteome	1200	0.126	0.112	+0.014	0.122
C_protperm	1	proteome	1200	0.029	0.016	+0.013	0.026
C_protperm	2	proteome	1200	0.105	0.112	−0.007	0.117
C_protperm	3	proteome	1200	0.088	0.071	+0.017	0.065
C_protperm	4	proteome	1200	0.083	0.090	−0.007	0.073
C_protperm	5	proteome	1200	0.083	0.120	−0.037	0.055
C_protperm	6	proteome	1199	0.067	0.073	−0.006	0.067
C_protperm	7	proteome	1199	0.043	0.040	+0.004	0.038
C_protperm	8	proteome	1199	0.045	0.037	+0.008	0.020
C_protperm	9	proteome	1199	0.083	0.066	+0.017	0.069
C_protperm	10	proteome	1199	0.050	0.050	+0.000	0.027
C_protperm	11	proteome	1199	0.073	–	–	0.028

Table 11: Collapse census of the small-trunk rounds. A run launches at the first epoch its validation prediction spread ratio reaches 0.05; a dead stretch is a maximal run of logged epochs after launch with the ratio below 0.01; a run collapsed if a dead stretch lasts 50 epochs or more, and never launched if no epoch reached 0.05 by epoch 200. Left: every dead stretch across both rounds by length, how many later returned to 0.05 and how many were still open at the run's last logged epoch. Right: runs per arm and launch cohort; v21's original cohort is the launch of 2026-10-06 that the in-place swap killed inside its jobs (pack 0 excepted), the relaunched cohort the attempt that ran to 1,200 epochs. Runs stopped by hand after collapsing keep their history. SRC: `experiments/019-simb-multimodal/scripts/collapse_census.py`

dead stretch, epochs	n	recovered	open
1 to 4	246	242	1
5 to 19	56	52	2
20 to 49	28	26	1
50 to 99	22	10	11
100 or more	42	8	31

round	arm	cohort	runs	collapsed	never launched
v21	`S_basis64`	original	12	3	0
v21	`S_basis64`	relaunched	12	3	0
v21	`S_hadam`	original	12	4	0
v21	`S_hadam`	relaunched	12	10	0
v21	`S_mask`	original	12	0	1
v21	`S_mask`	relaunched	12	0	0
v21	`S_nodrop`	original	12	2	0
v21	`S_nodrop`	relaunched	12	5	0
v21	`S_prop2`	original	12	0	0
v21	`S_prop2`	relaunched	12	0	0
v21	`S_prot`	original	12	0	0
v21	`S_prot`	relaunched	12	0	0
v21	`S_ref`	original	16	3	0
v21	`S_ref`	relaunched	8	1	0
v21	`S_sink`	original	12	1	0
v21	`S_sink`	relaunched	12	6	0
v21	`S_stack`	original	12	1	0
v21	`S_stack`	relaunched	12	4	0
v22	`F_b128`	original	3	2	0
v22	`F_b128lr1`	original	7	0	0
v22	`F_b128lr4`	original	3	0	3
v22	`F_b128wu`	original	3	0	0
v22	`F_b128wu_drop`	original	3	1	0
v22	`F_b128wu_hadam`	original	3	0	0
v22	`F_b128wu_wd`	original	3	0	0
v22	`F_b256lr1`	original	1	0	0
v22	`F_l4w180`	original	2	2	0
v22	`F_ref`	original	3	0	0

Table 12: Sequence baselines on the strains carrying both labels, the store v21 and v22 train on (1,103 train, 125 validation and 121 test strains on split seed 0; 6,169 expression and 1,850 proteome features). Pearson per feature across held-out strains, mean ± sd over 12 split seeds, with the ridge strength, gene rank and neighbor count selected on validation. The best validation mean per head is bold. SRC: `experiments/019-simb-multimodal/scripts/baselines_both_label_table.py`

head	baseline	embedding	validation Pearson	test Pearson
expression	bilinear ridge	ProtT5	0.090 ± 0.012	0.072 ± 0.030
expression	bilinear ridge	composite stack	0.071 ± 0.014	0.064 ± 0.015
expression	bilinear ridge	random 1024	0.019 ± 0.025	-0.006 ± 0.023
expression	neighbor average	ProtT5	**0.110 ± 0.017**	**0.090 ± 0.027**
expression	neighbor average	composite stack	0.064 ± 0.016	0.059 ± 0.018
expression	neighbor average	random 1024	0.017 ± 0.015	0.006 ± 0.014
proteome	bilinear ridge	ProtT5	0.063 ± 0.019	0.030 ± 0.033
proteome	bilinear ridge	composite stack	0.054 ± 0.016	0.027 ± 0.018
proteome	bilinear ridge	random 1024	0.019 ± 0.023	0.015 ± 0.021
proteome	neighbor average	ProtT5	**0.065 ± 0.013**	**0.040 ± 0.024**
proteome	neighbor average	composite stack	0.039 ± 0.011	0.026 ± 0.021
proteome	neighbor average	random 1024	0.009 ± 0.007	0.001 ± 0.008

11 2026-10-07, 19:00: the v21 round complete on twelve seeds ✗

The round ran to completion. Every one of the 104 relaunched runs reached epoch 1,199, at 42 to 45 s per epoch (56 s for the mask arm, whose schedule adds a masked pass), and the 26 jobs ended themselves between 10:01 and 14:35 on 2026-10-07, inside the limit. With pack 0's two survivors that is 106 finished runs over nine arms; the two missing `S_ref` splits (1 and 2) are the pack-0 runs stopped by hand after collapsing at epochs 1,102 and 939 and never relaunched. table 13 is the readout: per arm, the registered window (mean validation Pearson per feature over epochs 1,000 to 1,199) over every finished run and over the runs that did not collapse, and the paired difference against `S_ref` over the splits where neither run collapsed. Twelve splits resolve a paired gap of about 0.02; the healthy pairs are fewer, so their resolution is coarser still.

Table 13: The v21 small-trunk round: nine arms, twelve split seeds, 1,200 epochs at batch 32 on the both-label store. Window Pearson is the mean validation Pearson per feature over epochs 1,000 to 1,199 (the proteome head for `S_prot`, the expression head otherwise), mean ± sd over the finished runs and over the runs that did not collapse (a dead stretch of 50 or more epochs with the prediction spread ratio below 0.01). The paired column is the per-split difference against `S_ref`, mean and number of splits above zero, over the splits where neither run collapsed. The best healthy expression mean is bold. The sequence-baseline bar on this store is 0.110 (expression) and 0.065 (proteome), the neighbor average on ProtT5 (Table 12).
SRC: `experiments/019-simb-multimodal/scripts/v21_readout.py`

arm	change	finished	collapsed	window Pearson, all	window Pearson, healthy (n)	paired vs ref, healthy
`S_ref`	reference: single expression head, mask off	10/12	3	0.063 ± 0.037	0.070 ± 0.032 (9)	–
`S_mask`	mask schedule 0, 10, 100, 1000	12/12	0	0.094 ± 0.028	0.094 ± 0.028 (12)	+0.027 (7/9)
`S_sink`	trainable null sink in the perturbation head	12/12	6	0.052 ± 0.048	0.075 ± 0.030 (6)	+0.017 (5/5)
`S_prop2`	two-hop gated perturbation propagation	12/12	0	0.071 ± 0.044	0.071 ± 0.044 (12)	+0.010 (5/9)
`S_nodrop`	perturbation-head dropout 0	12/12	5	0.046 ± 0.047	0.079 ± 0.033 (7)	+0.001 (2/5)
`S_stack`	composite stack: FUDT, CaLM, ProtT5	12/12	4	0.070 ± 0.045	**0.095 ± 0.019 (8)**	+0.043 (6/6)
`S_basis64`	rank-64 response basis	12/12	3	0.054 ± 0.042	0.073 ± 0.032 (9)	+0.008 (5/7)
`S_hadam`	Hadamard pair term replacing the additive one	12/12	10	0.007 ± 0.014	0.018 ± 0.001 (2)	-0.040 (0/2)
`S_prot`	proteome head: protein abundance	12/12	0	0.066 ± 0.021	0.066 ± 0.021 (12)	–

Collapse decided the round more than any arm did. On the finished runs the collapse counts are: Hadamard 10 of 12, null sink 6, no dropout 5, composite stack 4, rank-64 basis 3, reference 3 of 12 (one of the eight relaunched), and the mask schedule, the two-hop propagation and the proteome head 0 of 12 each. The census over both rounds (table 11, regenerated) now holds 64 dead stretches of 50 epochs or more, of which 18 later returned to a spread of 0.05 (10 of the 22 between 50 and 99 epochs, 8 of the 42 at 100 or more); the gradient norm at onset is a median 1.15 times the preceding 50-epoch mean over the 64, unchanged from the morning's reading. In the table the original cohort's counts are low only because the swap killed those runs at epochs 270 to 930, before most onsets.

What the arms say. Three results stand on all twelve seeds, the rest are inside the noise.

- **The mask schedule is the one change that both removes collapse and raises the mean.** `S_mask` (schedule 0, 10, 100, 1,000) collapsed on 0 of 12 and scores 0.094 ± 0.028 over all twelve seeds against 0.063 ± 0.037 for the reference over its ten; paired, +0.033 over the ten shared splits (8 above zero) and +0.027 over the nine healthy pairs (7 above). It launches late, at epochs 123 to 284 against 39 to 125 for every other arm, which is the schedule withholding labels early. It is the best arm on 5 of the 12 splits; the reference is best on none.
- **The Hadamard operator is out.** `S_hadam` collapsed on 10 of 12 and its two survivors sit at 0.017 and 0.018, −0.040 against the reference on both healthy pairs and −0.055 over all ten shared splits (1 above zero). With the v22 warmup-arm reading (−0.031 on 3 of 3) that is two rounds and two machines against it.
- **The composite stack is the best model when it survives, and it survives two thirds of the time.** `S_stack` (FUDT, CaLM and ProtT5 together) scores 0.096 ± 0.019 over its eight healthy runs, +0.043 on 6 of 6 healthy pairs, with the highest eval-mode train Pearson of the round (median 0.65 against 0.52) and the widest prediction spread (0.44 against 0.35). It collapsed on 4 of 12, so over all twelve seeds it is 0.071 ± 0.046, level with the reference. Hypothesis (untested): the stack and the mask schedule compose, since one adds capacity and the other removes the collapse.

- The null sink (+0.017 on 5 of 5 healthy pairs, 6 of 12 collapsed), the rank-64 basis (+0.008, 5 of 7), the no-dropout arm (+0.002, 2 of 5, 5 collapsed) and the two-hop propagation (+0.010, 5 of 9) are within the twelve-seed resolution of the reference: not resolved, not null. Propagation collapsed on none but put 5 of 12 runs into a second low state that the collapse rule does not catch: spread 0.15 to 0.21, train Pearson 0.33 to 0.34, window −0.004 to 0.054, against 0.31 to 0.46 spread and 0.56 to 0.64 train Pearson on its other seven.
- **The proteome head sits at its sequence baseline.** `S_prot` scores 0.066 ± 0.021 over twelve seeds, 0 collapsed, against the twelve-seed neighbor-average bar of 0.065 on the same store (table 12); 6 of 12 runs are above it.

No arm mean clears the expression baseline. The twelve-seed neighbor average on ProtT5 scores 0.110 ± 0.017 on this store. Every arm mean is below it, the best (`S_mask`, `S_stack` healthy) by 0.015; 12 of the 94 finished expression runs are above it, the best at 0.146 (`S_nodrop` and `S_basis64` on split 6). The model's per-split scores vary by 0.03 to 0.05 across seeds within an arm, which is the same order as every contrast in the round except the Hadamard one.

Batch 128 on eleven seeds. IGB array 2426902 finished the batch-128 arm at the unscaled rate on split seeds 4 to 11 (three runs per A40, 15.4 s per epoch; 10.4 s with two on the card), which with the three GilaHyper seeds gives `F_b128lr1` on eleven seeds: window 0.051 to 0.141, mean 0.082 (table 8, regenerated), 0 of 11 collapsed. Paired against the batch-32 reference on the same split seeds (the GilaHyper `F_ref` on 0 to 2, the Delta `S_ref` on 4 to 11) it is +0.013 on 3 of 3 and +0.020 over 9 (6 above zero), +0.014 over the 8 pairs where the reference did not collapse (5 above): batch 128 at 3e-4 is level with or slightly above batch 32 at a quarter of the epoch time, and the eleven seeds do not resolve the sign. The batch-128 arm has now run 11 of 11 without collapse against the reference's 3 of 12 at batch 32 (hypothesis, untested as a contrast: the larger batch lowers the collapse rate).

v20 at eight partitions. Tasks 4 and 5 of the conditioned round finished at epoch 1,199 on `cabbi`; tasks 6 and 7 are at epochs 299 and 249, and tasks 8 to 11 wait under the two-task throttle. table 10 is regenerated on the eight. The proteome head conditioned on the strain's own expression is above the unconditioned arm at the same epochs on 8 of 8 (the six complete partitions by +0.022 to +0.079, mean +0.050; the two partial ones by +0.008 and +0.054) and above its permuted-expression control on 8 of 8. The expression head conditioned on the proteome is above the unconditioned arm on 7 of 8 (split 2 at −0.004; the complete six mean +0.024) and above its permuted control on 8 of 8. The reading of the morning stands: the gain is from the strain's own measurement, and both conditioned heads remain below the per-partition ridge from the measured other modality.